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**Evaluating the impact of a synthetic predator pheromone, a novel pest management tool, on target pests and insect communities in potato fields**

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**Abstract:**

Despite their reputation for species specificity, pheromones can influence interactions across trophic levels. This is particularly relevant in agricultural systems where predator-derived cues could influence pest and natural enemies. The predatory spined soldier bug, *Podisus maculiventris* (Say) (Hemiptera: Pentatomidae), preys upon the Colorado potato beetle, *Leptinotarsa decemlineata* (Say) (Coleoptera: Chrysomelidae), a major pest of solanaceous crops. Previous field work shows that a synthetic aggregation pheromone of *P. maculiventris* reduces beetle larval abundance through non-consumptive effects yet impacts on other herbivores and predators remain unclear. We hypothesized that synthetic predator pheromone cues alter herbivore and predator feeding behavior in an agroecosystem. We evaluated herbivore and natural enemy abundance and diversity in the field, measured predation using sentinel *L. decemlineata* egg clutches, and conducted greenhouse feeding assays with five predator taxa. Field results indicated a reduction in *L. decemlineata* larval abundance in pheromone-treated plots without detectable changes in overall herbivore or predator community abundance or diversity. Predation rates in pheromone-treated plots (mean  $\pm$  SE:  $9.8 \pm 2.9$  eggs consumed per three days) were five times higher compared to the control ( $1.4 \pm 0.5$  eggs consumed per three days). Greenhouse assays showed increased egg consumption by *P. maculiventris*. These findings suggest that the synthetic predator pheromone may enhance the foraging behavior of *P. maculiventris*, providing a potential mechanism for reduced *L. decemlineata* populations and a promising tool for potato pest management with minimal non-target impacts.

**Keywords:** chemical ecology, predator cues, non-consumptive effects, foraging behavior, behavioral control

## Introduction

Semiochemicals, especially sex pheromones, have become essential tools in agricultural pest management, with applications ranging from population monitoring and mass trapping to attract-and-kill, push–pull systems, and mating disruption. Their high specificity and relatively low environmental footprint make them attractive components of integrated pest management (IPM) programs (Sharma et al. 2019, Rizvi et al. 2021). For example, sex pheromones are widely deployed to disrupt mating in orchard pests such as the codling moth (*Cydia pomonella*), where pheromone-based disruption has reduced insecticide inputs and becomes a cornerstone of apple and pear IPM (Light et al. 2001, Witzgall et al. 2008). Along with sex pheromones, aggregation pheromones represent a promising new approach to manipulate pest populations because they attract and retain both sexes. A well-known example is the synthetic aggregation pheromone of the boll weevil (*Anthonomus grandis*), which has been widely used for monitoring and mass trapping in cotton production (Leonhardt et al. 1988). Similarly, aggregation pheromones have been successfully applied in other agricultural systems. An aggregation pheromone blend common across thrips has been used to manage greenhouse and field pests, including western flower thrips (*Frankliniella occidentalis*), through attract-and-kill strategies and behavioral interference (Kim et al. 2023). Another notable example that works in field settings is vittatalactone, an aggregation pheromone that successfully attracts both male and female striped cucumber beetles (*Acalymma vittatum*) (Weber et al. 2023).

However, the long-held assumption that pheromones act with strict species specificity is increasingly challenged with abundant evidence that pheromones influence interactions across multiple trophic levels. For instance, the bark beetle aggregation pheromone, ipsdienol, acts as an intraspecific attractant but also as a kairomone for its predator, the checkered beetle

(*Thanasimus dubius*) demonstrating how pheromone signals can be co-opted across trophic levels (Dahlsten et al. 2004). In addition, unlike sex pheromones, many aggregation pheromones are multicomponent blends produced and used by multiple taxa. These blends often include plant-derived volatiles or structurally similar convergent compounds, making them more likely to influence non-target organisms. For example, in the squash (*Cucurbita pepo*) system, squash bug (*Anasa tristis*) has been shown to eavesdrop on vittatalactone, the male-produced aggregation pheromone of the co-occurring striped cucumber beetle (*A. vittatum*) and increase its oviposition (Brzozowski et al. 2022). In another example, non-host green leaf volatiles (GLVs) from deciduous trees can also inhibit bark beetle pheromone responses, discouraging colonization of unsuitable hosts (Huber et al. 2021). Collectively, these examples demonstrate that semiochemicals often function through the community, influencing herbivores, predators, and parasitoids. As agroecosystems contain multiple interacting taxa, assessing these cross-level effects is essential for the ecologically safe and effective deployment of pheromone-based tools.

Pheromones and predator-associated volatiles can elicit strong non-consumptive effects (NCEs) on prey as individuals alter behavior to avoid predation. Exposure to predator odors can trigger dispersal, reduce feeding, and deter oviposition, ultimately suppressing pest populations. For example, aphids detect and avoid surfaces contaminated with chemical cues from lady beetles (Ninkovic et al. 2013) and reduce phloem ingestion in the presence of a lady beetle volatile cue (Kansman et al. 2023). Such behavioral shifts represent an underexplored yet promising avenue for pest suppression, where volatiles can deter pests and suppress feeding, leading to yield improvements (Martinez et al. 2025). In addition to directly influencing herbivore behavior, predator-associated cues can induce physiological and defensive responses in plants, which may further reduce herbivore performance through plant-mediated mechanisms.

However, as with other semiochemicals, predator cues may also affect non-target species. Predator-derived volatiles are likely to influence a wide array of arthropods, including beneficial predators and alternative prey, thereby reshaping community interactions. For example, Wen et al. (2023) found that a rove beetle pheromone spray reduced predatory spiders while overall arthropod diversity remained stable, indicating limited off-target effects. Still, many predatory insects respond to volatile cues that overlap chemically with herbivore-induced plant volatiles (HIPVs), which are broadly detectable by multiple organisms and can generate non-target and community-level effects (Kaplan 2012).

This study focuses on one such system involving the spined soldier bug, *Podisus maculiventris* (Say) (Hemiptera: Pentatomidae), a generalist predator and key natural enemy of the Colorado potato beetle (*Leptinotarsa decemlineata*) (Say) (Coleoptera: Chrysomelidae). Colorado potato beetle is among the most destructive insect pests of potato worldwide because of its capacity for rapid defoliation and widespread insecticide resistance (Alyokhin et al. 2022). Male *P. maculiventris* produces an aggregation pheromone that attracts both sexes and can modify prey responses, including reducing leaf consumption and larval abundance (Hermann and Thaler 2014, Aflitto and Thaler 2020). There are several hypotheses for why this pheromone may affect the community of herbivores or natural enemies in the system. Because this cue aggregates conspecifics, it may intensify intraspecific competition among naturally occurring or augmented *P. maculiventris* when prey are limited or alter intraguild interactions. *P. maculiventris* is known to act as an intraguild predator on the juveniles of other soft-bodied predators like lady beetles (Hough-Goldstein et al. 1996, Mallampalli et al. 2002). Conversely, herbivores may interpret the same cue as evidence of increased predation risk and adjust their behavior by reducing feeding, avoiding treated areas, or altering oviposition to lower the

likelihood of predator encounter (Werner and Peacor 2003, Hermann and Thaler 2014). The synthetic pheromone blend consists of  $\alpha$ -terpineol, terpinen-4-ol, benzyl alcohol, linalool, and (*E*)-2-hexenal (Aldrich et al. 1984). Two components, linalool and (*E*)-2-hexenal, are also green leaf volatiles released by potato plants following herbivore injury, and herbivore-induced plant volatiles (HIPVs) are broadly used by insects across taxa as indicators of herbivore activity, plant damage, and changes in the local risk landscape (Heil 2014). This chemical overlap between predator cues and plant volatiles suggests potential for community-level, non-target effects when the synthetic blend is deployed in the field.

In this study, we evaluate how a synthetic blend mimicking *P. maculiventris* aggregation pheromone influences the target pest, the Colorado potato beetle and the associated arthropod communities as well as enemy-pest interactions in potato fields. We hypothesized that the deployment of the predator pheromone would reduce Colorado potato beetle abundance, alter herbivore and predator communities, increase predation on Colorado potato beetle eggs, and influence herbivore and predator feeding behavior to *Podisus*-specific cues, either directly or indirectly through pheromone-induced changes in the abundance or behavior of naturally occurring predators. We used field experiments to test the effects of the synthetic predator pheromone on herbivore and natural enemy abundance and diversity using weekly D-vac sampling and visual counts. We tested the effects of pheromone on predation rates by deploying sentinel beetle egg clutches across the potato growing season. We conducted complementary greenhouse assays to test whether the pheromone affected predation rates of beetle eggs by dominant field predators. Understanding these dynamics is crucial for assessing the non-target and community-level impacts of deploying predator pheromones as a novel biocontrol tool in agroecosystems.

## Methods

### 1. Field experiments

#### *Field set-up*

On May 3, 2024, *Solanum tuberosum* (cv. Yukon Gold) potatoes were planted at Cornell University's Homer C. Thompson Vegetable Research Farm in Freeville, NY (42.517895, -76.332011). The variety Yukon Gold was selected due to its extensive use in prior research in this system (Aflitto and Thaler, 2020; Hermann and Thaler, 2014). Before planting, the field was fertilized with Triple 13 NPK at a rate of 1289 kg/ha. The experimental layout consisted of 36 single rows of potatoes, each row measuring 30 meters in length. Potato seeds were planted 0.3 m apart within rows, with a row spacing of 0.9 m. The field was organized into three blocks (A, B, and C), each containing 12 parallel rows. Blocks were separated by approximately four meters of bare soil and were oriented from right (A) to center (B) to left (C) of the site.

#### *Predator pheromone release*

The synthetic predator pheromone used in this study mimics the male aggregation pheromone of *P. maculiventris*, which is naturally released in spring to attract females. The dispenser containing the predator pheromone was made from a modified opaque brown 2 mL Eppendorf tube with three 0.55 mm holes in the top. Each dispenser was filled with 594.1 µL of an in-house blend of a synthetic male *P. maculiventris* dorsal abdominal gland pheromone mixture (354.0 µL (E)-2-hexenal, 216.0 µL alpha terpineol, 2.4 µL linalool, 2.0 µL terpinen-4-ol, and 19.7 µL benzyl alcohol) (Aldrich et al., 1984) and affixed to a highlighter green marking flag 20 cm high. Control plots received an empty tube attached to a marking flag. Pheromone dispensers have been shown to exert non-consumptive effects within approximately one meter of their placement (Aflitto and Thaler 2020; Martinez et al. 2025). To account for this, plots were designed to be

two meters long and included two pheromone dispensers per plot. The dispensers emitted the pheromone at an average rate of 0.23 mg/hr (Aflitto and Thaler 2020). This release rate was selected based on dispenser characterization described by Aflitto and Thaler (2020). Although individual *P. maculiventris* have been reported to produce up to approximately 1 mg of aggregation pheromone (Aldrich et al. 1984), natural pheromone exposure experienced by Colorado potato beetles in the field likely varies with predator density, spatial distribution, and environmental conditions. Consequently, our dispenser was intended to provide a biologically relevant pheromone stimulus rather than to replicate natural field concentrations exactly.

Pheromone treatments were deployed on May 30, 2024 to coincide with the emergence of overwintering Colorado potato beetle adults. Within each row, plots consisted of six plants spaced 30 cm apart in a single row, with two pheromone flags placed adjacent to the second and fourth plants. A five-meter buffer of untreated potato plants was maintained between plots. This spacing was selected based on previous field experiments in our lab indicating no effect on beetle abundance or feeding beyond approximately one meter from the pheromone dispensers and was considered sufficient to minimize potential pheromone movement among adjacent plots (Aflitto & Thaler, 2020; Martinez et. al. 2025). To further reduce side-by-side overlap of treatments, plots were staggered between rows.

Each row contained two pheromone-treated plots and two control plots. Pheromone-treated plots received two open-dispersion volatile dispensers, while control plots received two empty dispensers. Each plot type included two distinct sampling methods, as described below, and treatments were randomized across all rows for a total of 144 plots. Dispensers were left in the field throughout the measurement period and were replaced every 1.5 weeks.

### ***Field sampling***

We measured the effect of predator pheromone on the abundance of potato herbivores and natural enemies using two sampling methods. Starting June 4, 2024, and continuing weekly for seven weeks, we conducted visual counts of Colorado potato beetle predators, the Colorado potato beetle (eggs, larvae, and adults), and other potato pests present on the six plants within each of the 72 plots (half control, half with pheromone). Colorado potato beetle predators were identified based on published literature describing established natural enemies of CPB (Weber et al. 2022). Insects were identified visually in the field and were not collected during visual surveys. For each plant, the observers positioned themselves approximately 15–30 cm from the plant material and conducted a systematic inspection. Observers gently manipulated the stems and foliage to examine both the upper and lower surfaces of all leaves, including those near the base of the plant, to ensure full coverage of the plant canopy. To minimize potential observer bias, observers were rotated among plots during each sampling period so that no observer consistently sampled the same plots. In addition, all observers had prior entomological experience and received additional training before sampling on the identification of arthropod taxa commonly encountered in potato fields. An identification guide was available during each sampling event, and a trained entomologist was available to verify uncertain field identifications.

In addition to weekly visual estimates, we used a handheld D-vac device (Model 122 Rincon-Vitova Insectaries, Ventura CA) on the other 72 plots (half control, half with pheromone). Each plant was vacuumed for 5–10 seconds. Samples from all six plants were placed into a collection bag labeled with block, plot ID, row, date of collection and placed in the freezer.

### ***Insect Identification***

All observed organisms were identified to family, with taxa of particular importance for Colorado potato beetle predation and potato pest status identified to finer taxonomic resolution when possible. Focal herbivores included the potato leafhopper, *Empoasca fabae* (Harris) (Hemiptera: Cicadellidae); the Colorado potato beetle, *Leptinotarsa decemlineata* (Say) (Coleoptera: Chrysomelidae); aphids (Hemiptera: Aphididae); the tarnished plant bug, *Lygus lineolaris* (Palisot de Beauvois) (Hemiptera: Miridae); and flea beetles (Coleoptera: Chrysomelidae: Galerucinae) (Weber et al. 2022). Predators were identified to family and included lady beetles (Coleoptera: Coccinellidae), minute pirate bugs (Hemiptera: Anthocoridae), spiders (Araneae: Araneidae), damsel bugs (Hemiptera: Nabidae), and lacewings (Neuroptera: Chrysopidae, Hemerobiidae). Species-level identification was conducted for key predators of the Colorado potato beetle, including *Lebia grandis* (Coleoptera: Carabidae) (Hentz) and *Podisus maculiventris* (Hemiptera: Pentatomidae) (Say). Lady beetles were further identified to *Coleomegilla maculata* (Coleoptera: Coccinellidae) (De Geer), a native, naturally occurring predator of Colorado potato beetle, whereas other coccinellids were retained at the family level for community analysis. Herbivore and predator taxa representing less than 1% of total observations were not identified further, as these uncommon taxa were unlikely to play a substantial role in pest suppression or plant damage within this system (Chang and Snyder 2004). Although excluding rare taxa may slightly underestimate richness and diversity, these taxa occurred at very low frequencies and were not expected to substantially influence community-level patterns or conclusions.

### ***Egg predation rates in the field***

To quantify Colorado potato beetle egg predation rates in the field over the potato growing season, we deployed sentinel beetle eggs weekly from June 7 to July 19, 2024. We prepared sentinel eggs

using three-week-old potted potato plants grown in the greenhouse and fresh egg clutches (20 to 40 eggs per clutch) collected from a lab colony of the Colorado potato beetle. We used a paired design where an egg clutch was glued to the lower side of a leaf (the “predator-exposed” clutch) and another clutch was glued to the lower side of a nearby leaf of the same plant enclosed in a mesh bag (the “control” clutch). Each week, we randomly selected five to seven control and pheromone-treated plots across our study site and deployed one plant with sentinel eggs in each plot for three days (no repeated deployments in the same locations over the season). The number of lost eggs in each clutch was determined as the difference between the initial number of eggs and the number of remaining intact eggs (partially damaged and shrunk eggs were counted as lost eggs as they were not able to hatch). To account for non-predation egg mortality, background egg loss from the paired control clutch was scaled to the size of the predator-exposed clutch and subtracted from observed egg loss to estimate predation. The average non-predation mortality was low and similar in the control and pheromone-treated plots (mean  $\pm$  SE; control plot:  $0.77 \pm 0.23$  eggs per three days; pheromone-treated plot:  $0.75 \pm 0.27$  eggs per three days). Observations with clutches (either control or predator-exposed) that were missing or hatched were excluded, giving a final sample size of 50 (30 in the control plots vs. 20 in the pheromone-treated plots) in the analysis.

## 2. Lab experiments

### *Predation assays*

Greenhouse feeding assays were conducted using the dominant Colorado potato beetle egg predators that were also abundant in the field, focusing on the species that could be reliably collected or maintained under laboratory conditions. These included *Coleomegilla maculata* adults and fourth-instar larvae, *Lebia grandis* adults, *Chrysoperla rufilabris* third-instar larvae,

228 and *P. maculiventris* adults. The assays took place in the greenhouse facility at Cornell  
229 University in Ithaca, NY. Minute pirate bug *Orius insidiosus* was also assessed through this  
230 study due to its high abundance in the field, however there was no clear consumption evidence of  
231 eggs, and no further analysis was performed. *Coleomegilla maculata* adults, *Lebia grandis* and  
232 *P. maculiventris* adults were field collected from experimental plots located in Freeville, NY in  
233 August 2024. *Chrysoperla rufilabris* and *Orius insidiosus* were purchased from ARBICO  
234 Organics, Oro Valley, AZ. Additional adult *Coleomegilla maculata* and all fourth-instar  
235 *Coleomegilla maculata* larvae were taken from our lab-colony reared in a growth chamber  
236 maintained on a 16:8 L:D dark cycle and fed on pea aphids.

237 For the greenhouse assays, predators were randomly selected from the lab colony once predators  
238 started reproducing or assays were done following field collection. Prior to the experiment,  
239 predators were isolated individually without food for 24 hours to standardize hunger levels.

240 Each replicate contained a single potato plant with a set number of *Leptinotarsa decemlineata*  
241 eggs affixed to the abaxial side of a single, mid-height leaf blade. The beetle eggs were collected  
242 from lab-maintained colonies in a growth chamber on a 16:8 L:D dark cycle. Temperature was  
243 maintained at approximately 28.2°C. All eggs were collected within 24 hours before the start of  
244 the assays. Plants were spaced equally in the greenhouse, about 0.5 meters apart. Individual  
245 predators were introduced to the plants randomly assigned to either the control or the treatment  
246 group. Plants in the control group were provided rubber septas charged with 250 microliters of  
247 hexane solution and soaked overnight. Plants in the treatment group had rubber septas charged  
248 with 250 microliters of 1% pheromone solution (see above) diluted with hexane, attached to a  
249 leaf around 10-15 cm across from the egg-affixed leaf. All plants were covered by perforated  
250 bread bags and secured at the base of the plant with a rubber band. Replicates were evenly

divided between control and pheromone treatments. Due to space constraints, the experiments were carried out over sequential trials, each containing an equal number of control and pheromone replicates. Total eggs consumed were counted after 24 or 48 hours. Predators that died or pupated during the assay were omitted from analysis.

The predation assays were done with *C. maculata* adults (n = 128; 63 control, 65 pheromone), *C. maculata* fourth-instar larvae (n = 27; 15 control and 12 pheromone), *L. grandis* adults (n = 32; 16 control and 16 pheromone), *C. rufilabris* third-instar larvae (n = 46; 23 control and 23 pheromone), and *P. maculiventris* adults (n = 62; 31 control and 31 pheromone; this assay was conducted for 48 hours).

### 3. Data analysis

#### *Abundance of the Colorado potato beetle*

We analyzed the abundance of the Colorado potato beetle (*L. decemlineata*) separately for the following biological and statistical reasons: (1) It is the focal pest species in this system. (2) It has known behavioral responses to predator-derived cues (Aflitto and Thaler, 2020; Helms et al. 2019). (3) It was not found in D-vac samples. (4) Its statistical distribution differed from that of other herbivores in the system. We fit a generalized linear mixed-effects model with beetle larval abundance as the response variable, pheromone treatment as the fixed effect, with week and plot nested within the row and block as the random effects.

#### *Abundance, richness, and diversity of arthropod communities*

To evaluate how pheromone treatment influenced herbivore and predator community structure, we calculated total abundance, mean abundance per taxonomic group, richness, and Shannon diversity for each field sample. Herbivore and predator datasets were first filtered to include only

non-predatory taxa or predatory taxa only, and total counts were summed within each combination of block, row, plot, treatment, sampling method, week, and taxonomic group. Community metrics were calculated at the level of taxonomic groups (family-level units), including taxonomic richness (number of families detected), mean abundance per family, and Shannon diversity using the “vegan” package (Oksanen, 2024). Although community metrics were calculated at the family level, which may mask species-specific responses within some groups, potato arthropod communities are typically dominated by a relatively small number of herbivore and predator taxa. Consequently, we expect family-level metrics to capture the dominant community responses to pheromone deployment, while recognizing that species-level responses within families may have gone undetected.

Abundance was log-transformed to meet model assumptions and analyzed with a linear mixed-effects model including treatment as the fixed effect and week and plot nested within block as the random effects. Richness was modeled using a generalized linear mixed model with a Poisson error distribution, with treatment included as the fixed effect and week and plot nested within block as the random effects. This model was fitted via the `glmmTMB()` function in the R “glmmTMB” package (Brooks et al. 2017). The effects of treatment on Shannon diversity were evaluated using a linear mixed-effects model with treatment as the fixed effect and week and plot nested within block as the random effects. We checked model assumptions using the simulated scaled residuals generated from the `simulateResiduals()` function in the R “DHARMa” package (Hartig 2024).

### *Abundance of selected predators and herbivores*

To evaluate the effects of pheromone treatment on herbivorous and predator arthropod abundance across taxonomic groups, we constructed generalized linear mixed models using the

lme4 package (Bates et al. 2015). Herbivore and predator datasets were filtered to include taxa that were consistently detected across sampling events. The selected herbivores included *Cicadellidae* (potato leaf hopper), *Galerucinae* (flea beetles), *Miridae* (tarnished plant bug), and *Aphididae* (Aphids); the selected predators included *P. maculiventris*, Neuroptera (lacewings), Coccinellidae (lady beetles), Nabidae (damselflies), Araneidae (spiders), and Anthrocoridae (minute pirate bugs). Abundance for each taxonomic group was calculated as the mean number of individuals collected within each combination of block, plot, row, treatment, sampling method, and week.

We modeled herbivore abundance as a function of treatment, taxonomic group, and their interaction using a negative binomial GLMM. The full model included the interaction between treatment and taxonomic group as fixed effects, with plot nested within block and week included as random effects. A negative binomial error distribution was used to account for overdispersion in count data. Post hoc comparisons among treatment levels within each taxonomic group were performed using pairwise contrasts of estimated marginal means (EMMs) with the emmeans package (Lenth 2025).

### ***Egg predation in the field***

To compare the egg predation rates between the control and the pheromone-treated plots, we fit a generalized linear model with the number of predated eggs as the response, and plot treatment type and week as the predictors. We used a negative binomial distribution with a log link function to account for data overdispersion. The model was fitted via the glmmTMB() function in the R “glmmTMB” package (Brooks et al. 2017). We checked model assumptions using the simulated scaled residuals generated from the simulateResiduals() function in the R “DHARMa” package (Hartig 2024). We used the Wald chi-square test to assess predictor significance using the Anova()

function (type II sums of squares) in the R “car” package (Fox & Weisberg 2018). The analysis was performed in R version 4.3.3 (R Core Team 2024).

### ***Egg predation in the greenhouse***

We analyzed the probability of egg consumption by four predator species (*P. maculiventris*, *C. maculata*, *L. grandis*, and *C. rufilabris*) using a generalized linear mixed model fitted with the glmmTMB package (Brooks et al. 2017). The response variable was the number of eggs consumed relative to the number available at the start of the assay, modeled as a binomial response. Fixed effects included treatment, predator life stage/species, their interaction, and assay length as a covariate. Random effects included trial identity since experiments were conducted sequentially, and plant identity. The species *C. maculata* was separated to analyze the treatment effect on egg consumption in larvae and adult stages. *Coleomegilla maculata* was selected because it represented the highest proportion of lady beetles observed in the field, while *L. grandis* was included due to its role as an important specialist predator of the Colorado potato beetles. All analysis was conducted in R version 4.4.0 (R Core Team 2025).

## **Results**

### ***Herbivore abundance and community structure***

While the abundance of the target herbivore, the Colorado potato beetle, was reduced by the pheromone, none of the other herbivores were affected. The estimated mean abundance of Colorado potato beetle larvae was 32% lower in pheromone-treated plots than in control plots (5.53 vs. 8.08 larvae per plot;  $z = -2.51$ ,  $P = 0.01$ ; Fig. 1).

The presence of the synthetic predator pheromone did not significantly alter herbivore abundance, richness, or diversity of the rest of the herbivore community over the seven-week

sampling period (Appendix S1; Fig. S1; Table S1). The herbivore community was dominated by potato leafhoppers, which comprised approximately 79% of all herbivores observed, followed by the Colorado potato beetles (6%), aphids (5%), tarnished plant bugs (5%), and flea beetles (4%). Together, these taxa accounted for approximately 99% of the herbivore community.

Aphids, potato leafhoppers, flea beetles, and tarnished plant bugs did not differ significantly in abundance between the control and pheromone treatments (Fig. 2; Table 1).

### ***Predator community structure***

Exposure to the synthetic predator pheromone did not significantly alter predator abundance, richness, or diversity over the seven-week sampling period (Appendix S1; Fig. S2; Table S2). The predator community was dominated by lady beetles, which comprised approximately 48% of all predators observed, followed by minute pirate bugs (22%), spiders (12%), damsel bugs (6%), spined soldier bugs (5%), and lacewings (4%). Together, these taxa accounted for approximately 97% of the predator community.

The most abundant predators in the potato field—minute pirate bugs, spiders, lady beetles, damsel bugs, lacewings, and spined soldier bugs, did not differ significantly in abundance between the control and pheromone treatments (Fig. 3; Table 2).

### ***Predation rates of Colorado potato beetle eggs in the field***

Overall egg predation rates over a two-month period were higher in the pheromone-treated plots (mean [95% CI]: 9.8 [2.7–15.9] eggs per three days) compared to the control plots (mean [95% CI]: 1.4 [0.4–2.4] eggs per three days) ( $\chi^2 = 11.7$ ;  $df = 1$ ;  $p < 0.001$ ; Fig. 4).

### **Predation of Colorado potato beetle eggs in greenhouse assays**

There is no significant effect of pheromone treatment on the proportion of eggs consumed by adult *C. maculata* (81.9% vs. 82.9%;  $Z = 0.09$ ;  $p = 0.92$ ; Fig. 5), *C. maculata* larvae (85.5% vs. 68.3%;  $Z = 0.83$ ;  $p = 0.41$ ); *C. rufilabris* (20.7% vs. 13.9%;  $Z = 0.52$ ;  $p = 0.60$ ; Fig. 5), or *L. grandis* (31.1% vs. 66.1%;  $Z = -1.30$ ;  $p = 0.19$ ; Fig. 5).

There is a significant effect of pheromone treatment on the proportion of eggs consumed by *P. maculiventris* ( $Z = -3.69$ ;  $p < 0.001$ ; Fig. 5) with estimated egg consumption increasing from 1.3% in the control treatment to 29.2% in the pheromone treatment.

### **Discussion**

While the abundance of the target herbivore (the Colorado potato beetle) was reduced by exposure to the synthetic predator pheromone, the abundance, species richness, and diversity of herbivores and natural enemies in the field were not altered throughout the season (Appendix S1; Fig. S3-S4). Despite these neutral community responses, predation on beetle eggs was significantly higher in pheromone-treated plots, suggesting that the pheromone may have influenced predator foraging behavior rather than predator abundance. However, predator behavior was not directly measured in this study. Greenhouse assays further revealed that *P. maculiventris* increased its feeding rate in the presence of the pheromone, demonstrating that the compound can enhance consumptive effects for at least one key predator species. In contrast, *C. maculata* lady beetle larvae and adults, lacewing larvae, and *L. grandis* showed no detectable response. Collectively, although predators varied in their feeding responses to the synthetic pheromone, the net outcome positively favored future use of the pheromone in biological control of the Colorado potato beetle.

The lack of community-wide responses by herbivores may reflect differences in how taxa perceive or respond to predator-associated volatiles. Most herbivores in this system, aside from the Colorado potato beetle, are not known to be preyed upon by *P. maculiventris* and therefore may have little incentive to respond directly to its cues. Although previous work demonstrates that predator pheromones can alter plant physiology and quality, both in potatoes exposed to the *P. maculiventris* pheromone (Aflitto and Thaler, 2020; Martinez et al. 2025) and in other systems where plants respond to predator-derived cues (Helms et al. 2017), the herbivores in our study did not show detectable changes in a way that suggested response to plant quality. Consistent with this, the Colorado potato beetle responded to predator-pheromone-induced changes in plant quality in previous greenhouse experiments (Martinez et al. 2025), but we found no evidence that other herbivores reacted similarly in our field surveys. Many chewing herbivores adjust behavior rather than population numbers in response to predator cues (Kaplan et al. 2014), though such subtle shifts may have gone undetected as we did not measure behavior. Additionally, herbivores such as leafhoppers and aphids rely heavily on plant nutritional quality and phloem chemistry, traits that may not be strongly influenced by exposure to the predator pheromone (Hewer et al. 2010). Chewing herbivores like flea beetles, despite sharing a feeding mode with Colorado potato beetle larvae, may also differ in how they perceive and respond to predator-associated volatiles (Li et al. 2024).

Aggregation pheromones have been shown to attract adult *P. maculiventris* in field settings, yet such recruitment was not detected in our experiment (Kelly et. al. 2014). Predators may vary widely in their responses to the pheromone. Notably, *C. maculata* has been shown to respond to a component of the *P. maculiventris* pheromone under field conditions (Zhu et al. 1999; Kaplan 2012), suggesting capacity for attraction in at least some members of the predator

407 guild. There is evidence that predators can be attracted by HIPVs in field settings; for example,  
408 methyl salicylate reliably pulls in diverse predator guilds, showing that single semiochemicals  
409 can serve as attractive cues (James, 2005; Zhang et al., 2006; Zhu and Park, 2005). Additionally,  
410 our D-vac and visual sampling methods quantified predator abundance rather than predator  
411 activity and may not have detected localized changes in predator behavior within plots.  
412 However, predator abundance in our fields remained unchanged across treatments, and predators  
413 did not appear to increase their presence in plots with higher Colorado potato beetle larval  
414 abundance. This indicates that the pheromone was neither a deterrent nor an attractive cue at the  
415 plot scale, nor were any patches of larvae serving as a prey hotspot. Predation rates on Colorado  
416 potato beetle eggs changed in the field, prompting us to examine predator feeding behavior in  
417 greenhouse assays, where we did observe species-specific responses to the pheromone. *Podisus*  
418 *maculiventris* increased feeding, which may be due to the species' established use of aggregation  
419 pheromones in mate attraction and prey sharing (Sant'Ana et al. 1997). Alternatively, increased  
420 egg consumption may reflect a behavioral response to aggregation cues that elevate local  
421 predator density, as higher conspecific densities commonly alter per-capita feeding rates through  
422 interference or competitive interactions in biological control systems (DeLong and Vasseur,  
423 2011). However, this mechanism was not directly tested in our study and warrants future  
424 investigation. Although not statistically significant, *L. grandis* exhibited a moderate increase in  
425 egg consumption under pheromone exposure, suggesting potential responsiveness among some  
426 beetle predators. Not all predators exhibited behavioral responses to the pheromone treatment. *C.*  
427 *maculata* adults did not respond to the predator pheromone. While there is evidence of adult  
428 response to an individual component of the pheromone (Zhu et al. 1999; Kaplan 2012) this  
429 sensitivity may not translate into attraction or altered feeding behavior in response to a more

complex pheromone blend. Similarly, *C. maculata* larvae and lacewings showed no significant differences in feeding rates, suggesting that the pheromone does not universally enhance predation across the natural enemy guild. This lack of response is noteworthy given that these larval predators could be at risk of intraguild predation by adult stink bugs (Abram et al. 2014; Mallampalli et al. 2002). Predator-derived cues have been shown to trigger behavioral and physiological responses in insect prey, though whether such effects consistently scale up to influence population abundance or community structure remains poorly resolved (Hermann and Landis, 2017). Such asymmetries highlight the importance of considering functional traits and ecological roles of the larger community when evaluating the potential of semiochemicals as biocontrol tools.

From an applied perspective, our findings demonstrate that predator-derived pheromones can enhance biological control. By increasing predation on Colorado potato beetle eggs, the pheromone has the potential to improve the biocontrol efficiency of key predators, particularly given that Colorado potato beetle feeding damage can reduce potato yield when defoliation exceeds the plants compensatory capacity (Hare, 1990; Stieha and Poveda, 2015). The low numbers of *P. maculiventris* observed in the field suggest that the increased predation in pheromone-treated plots likely resulted from subtle shifts in predator behavior rather than from increases in predator abundance. This is notable because *P. maculiventris* is difficult to retain in cropping systems and often requires multiple augmentative releases (Aldrich and Cantelo, 1999; Biever and Chauvin, 1992; Tipping et al. 1999); even a small increase in conspecific attraction, combined with higher per-capita feeding rates, could meaningfully improve its efficacy. Importantly, this enhancement occurred without detectable reductions in the abundance of either beneficial insects or non-target herbivores, indicating that pheromone deployment is unlikely to

generate the unintended impacts commonly associated with broad-spectrum insecticides. Although these findings demonstrate the potential of predator pheromones for enhancing biological control, our conclusions are based on a single growing season at one location in a potato production system using a single pheromone release rate. Future studies evaluating multiple environments, cropping systems, years, and pheromone release rates will be important to determine the generality and optimize the application of this approach. Collectively, these outcomes support the integration of predator pheromones into push–pull or attract-and-stimulate strategies to strengthen pest suppression in potato systems.

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- Figure 1:** Mean ( $\pm$  SE) number of Colorado potato beetle larvae per plot in control and pheromone-treated plots averaged across the sampling period. Raw observed means were  $8.51 \pm 0.89$  larvae per plot in the control treatment ( $n = 177$  observations) and  $6.73 \pm 0.82$  larvae per

616 plot in the pheromone treatment ( $n = 159$  observations). Error bars represent  $\pm 1$  SE. Asterisk  
617 indicates a significant treatment effect ( $p < 0.05$ ).

618 **Alt text 1:** Bar plot showing number of Colorado potato beetle larvae in the control (yellow bar)  
619 and the predator pheromone (purple bar) plots. There are less beetles in the plots with the  
620 predator pheromone.

621 **Figure 2.** Average herbivore number per plot across two months in control and pheromone-  
622 treated plots. Herbivores are shown by family and were the major herbivore families found in the  
623 field. Aphididae (aphids), Cicadellidae (potato leafhopper), Galerucinae (flea beetles), and  
624 Miridae (tarnished plant bugs). Error bars represent standard errors.

625 **Alt text 2:** Number of herbivores in each plot (control or pheromone). Listed by different  
626 families. From right to left – Aphididae, Cicadellidae, Galerucinae, Miridae.

627 **Figure 3.** Average predator number per plot across two months in control and pheromone treated  
628 plots. Predators are shown by family and are the most abundant in the field, Anthocoridae,  
629 Araneidae, Neuroptera (Chrysopidae/Hemerobiidae), Coccinellidae, Nabidae, and Pentatomidae.  
630 Error bars represent standard errors.

631 **Alt text 3:** Number of predators in each plot (control or pheromone). Listed by different  
632 families. From right to left – Anthocoridae, Araneidae, Chrysopidae/Hemerobiidae,  
633 Coccinellidae, Nabidae, Pentatomidae.

634 **Figure 4.**

635 Overall beetle egg predation rates in the control (mean [95% CI]: 1.4 [0.4–2.4]) and the  
636 pheromone-treated plots (mean [95% CI]: 9.8 [2.7–15.9]) over a two-month period. Error bars

represent standard errors. Asterisk denotes a significant difference between the treatment levels ( $p < 0.05$ ).

**Alt text 4:** Predation rates of predators found in potato field. Rates represent eggs consumed every three days in control or pheromone treated plots.

**Figure 5.** Percent of the Colorado potato beetle eggs consumed by five predator species/life stages when exposed to predator pheromone or control volatiles in greenhouse assays. Panels from left to right show *Coleomegilla maculata* adults, *Chrysoperla rufilabris*, *Lebia grandis*, *Podisus maculiventris*, and *Coleomegilla maculata* larvae. Error bars represent standard errors. Asterisks denote significant differences between treatment levels ( $p < 0.05$ ).

**Alt text 5:** Percent of Colorado potato beetle eggs consumed by predators in greenhouse assays. From right to left – *Coleomegilla maculata* adults, *Coleomegilla maculata* larvae, *Lebia grandis*, *Chrysoperla rufilabris*, *Podisus maculiventris*.

| Taxonomic group     | Control /<br>Pheromone ratio | SE   | Z     | p-value |
|---------------------|------------------------------|------|-------|---------|
| <i>Aphididae</i>    | 1.10                         | 0.09 | 1.09  | 0.28    |
| <i>Cicadellidae</i> | 1.07                         | 0.07 | 1.04  | 0.29    |
| <i>Galerucinae</i>  | 0.93                         | 0.09 | -0.72 | 0.47    |
| <i>Miridae</i>      | 1.10                         | 0.09 | 0.14  | 0.89    |

**Table 1.** Control /pheromone abundance ratios ( $\pm$  SE) for dominant herbivore taxa. Ratios were estimated from generalized linear mixed models; values significantly different from 1 would indicate treatment effects. No significant differences were detected ( $p < 0.05$ ).

| Taxonomic Group      | Control /<br>Pheromone ratio | SE   | Z     | <i>p</i> -value |
|----------------------|------------------------------|------|-------|-----------------|
| <i>Anthocoridae</i>  | 1.11                         | 0.17 | 0.69  | 0.49            |
| <i>Araneidae</i>     | 0.81                         | 0.25 | -0.68 | 0.50            |
| <i>Coccinellidae</i> | 0.99                         | 0.09 | -0.14 | 0.89            |
| <i>Nabidae</i>       | 0.92                         | 0.24 | -0.33 | 0.74            |
| <i>Neuroptera</i>    | 1.12                         | 0.35 | 0.46  | 0.65            |
| <i>Podisus</i>       | 1.11                         | 0.38 | 0.36  | 0.76            |

**Table 2.** Control/pheromone abundance ratios ( $\pm$  SE) for dominant predator taxa. Ratios were estimated from generalized linear mixed models; values significantly different from 1 would indicate treatment effects. No significant differences were detected ( $p < 0.05$ ).

## Abstract:

Despite their reputation for species specificity, pheromones can influence interactions across trophic levels. This is particularly relevant in agricultural systems where predator-derived cues could influence pest and natural enemies. The predatory spined soldier bug, *Podisus maculiventris* (Say) (Hemiptera: Pentatomidae), preys upon the Colorado potato beetle, *Leptinotarsa decemlineata* (Say) (Coleoptera: Chrysomelidae), a major pest of solanaceous crops. Previous field work shows that a synthetic aggregation pheromone of *P. maculiventris* reduces beetle larval abundance through non-consumptive effects, yet ~~potential~~ impacts on other herbivores and predators remain unclear. We hypothesized that synthetic predator pheromone cues ~~could~~ alter herbivore and predator feeding behavior in an agroecosystem. We evaluated herbivore and natural enemy abundance and diversity in the field, measured predation using sentinel *L. decemlineata* egg clutches, and conducted greenhouse feeding assays with five predator taxa. ~~A field experiment evaluated the effects of the synthetic predator pheromone on herbivore and natural enemy abundance and diversity. We also deployed sentinel *L. decemlineata* egg clutches across the season to measure field predation rates. Greenhouse feeding assays tested whether common predators altered consumption of *L. decemlineata* eggs in response to the pheromone.~~ Field results indicated a reduction in *L. decemlineata* larval abundance in pheromone-treated plots without detectable changes in overall herbivore or predator community abundance or diversity. Predation rates in pheromone-treated plots (mean  $\pm$  SE:  $9.8 \pm 2.9$  eggs consumed per three days) were five times higher compared to the control ( $1.4 \pm 0.5$  eggs consumed per three days). ~~However, predation rates in pheromone-treated plots were five times higher compared to the control.~~ Greenhouse assays showed increased egg consumption by *P. maculiventris*. These findings suggest that the synthetic predator pheromone

may ~~enhance the~~ ~~reduce *L. decemlineata* populations by enhancing~~ foraging behavior of *P. maculiventris*, providing a potential mechanism for reduced *L. decemlineata* populations and a promising ~~providing a potential~~ tool for potato pest management with minimal non-target impacts. ~~on herbivore and predator communities.~~

**Keywords:** chemical ecology, predator cues, non-consumptive effects, foraging behavior, behavioral control

## Introduction

Semiochemicals, especially sex pheromones, have become essential tools in agricultural pest management, with applications ranging from population monitoring and mass trapping to attract-and-kill, push–pull systems, and mating disruption. Their high specificity and relatively low environmental footprint make them attractive components of integrated pest management (IPM) programs (Sharma et al. 2019, Rizvi et al. 2021). For example, sex pheromones are widely deployed to disrupt mating in orchard pests such as the codling moth (*Cydia pomonella*), where pheromone-based disruption has reduced insecticide inputs and becomes a cornerstone of apple and pear IPM (Light et al. 2001, Witzgall et al. 2008). Along with sex pheromones, aggregation pheromones represent a promising new approach to manipulate pest populations because they attract and retain both sexes. A well-known example is the synthetic aggregation pheromone of the boll weevil (*Anthonomus grandis*), which has been widely used for monitoring and mass trapping in cotton production (Leonhardt et al. 1988). Similarly, aggregation pheromones have been successfully applied in other agricultural systems. ~~For instance, an~~ An aggregation pheromone blend common across thrips has been used to manage greenhouse and field pests, including western flower thrips (*Frankliniella occidentalis*), through attract-and-kill strategies and behavioral interference (Kim et al. 2023). Another notable example that works in field

settings is vittatalactone, an aggregation pheromone that successfully attracts both male and female striped cucumber beetles (*Acalymma vittatum*)(Weber et al. 2023).

However, the long-held assumption that pheromones act with strict species specificity is increasingly challenged with abundant evidence that pheromones influence interactions across multiple trophic levels. For instance, the bark beetle aggregation pheromone, ipsdienol, acts as an intraspecific attractant but also as a kairomone for its predator, the checkered beetle (*Thanasimus dubius*) demonstrating how pheromone signals can be co-opted across trophic levels (Dahlsten et al. 2004). In addition, unlike sex pheromones, many aggregation pheromones are multicomponent blends produced and used by multiple taxa. These blends often include plant-derived volatiles or structurally similar convergent compounds, making them more likely to influence non-target organisms. For example, in the squash (*Cucurbita pepo*) system, squash bug (*Anasa tristis*) has been shown to eavesdrop on vittatalactone, the male-produced aggregation pheromone of the co-occurring striped cucumber beetle (*A. vittatum*) and increase its oviposition (Brzozowski et al. 2022). In another example, non-host green leaf volatiles (GLVs) from deciduous trees can also inhibit bark beetle pheromone responses, discouraging colonization of unsuitable hosts (Huber et al. 2021). Collectively, these examples demonstrate that semiochemicals often function through the community, influencing herbivores, predators, and parasitoids. As agroecosystems contain multiple interacting taxa, assessing these cross-level effects is essential for the ecologically safe and effective deployment of pheromone-based tools.

Pheromones and predator-associated volatiles can elicit strong non-consumptive effects (NCEs) on prey as individuals alter behavior to avoid predation. Exposure to predator odors can trigger dispersal, reduce feeding, and deter oviposition, ultimately suppressing pest populations. For example, aphids detect and avoid surfaces contaminated with chemical cues from lady

beetles (Ninkovic et al. 2013) and reduce phloem ingestion in the presence of a lady beetle volatile cue (Kansman et al. 2023). Such behavioral shifts represent an underexplored yet promising avenue for pest suppression, where volatiles can deter pests and suppress feeding, leading to yield improvements (Martinez et al. 2025). In addition to directly influencing herbivore behavior, predator-associated cues can induce physiological and defensive responses in plants, which may further reduce herbivore performance through plant-mediated mechanisms. However, as with other semiochemicals, predator cues may also affect non-target species. Predator-derived volatiles are likely to influence a wide array of arthropods, including beneficial predators and alternative prey, thereby reshaping community interactions. For example, Wen et al. (2023) found that a rove beetle pheromone spray reduced predatory spiders while overall arthropod diversity remained stable, indicating limited off-target effects. Still, many predatory insects respond to volatile cues that overlap chemically with herbivore-induced plant volatiles (HIPVs), which are broadly detectable by multiple organisms and can generate non-target and community-level effects (Kaplan 2012).

This study focuses on one such system involving the spined soldier bug, *Podisus maculiventris* (Say) (Hemiptera: Pentatomidae), a generalist predator and key natural enemy of the Colorado potato beetle (*Leptinotarsa decemlineata*) (Say) (Coleoptera: Chrysomelidae). Colorado potato beetle is among the most destructive insect pests of potato worldwide because of its capacity for rapid defoliation and widespread insecticide resistance (Alyokhin et al. 2022). Male *P. maculiventris* produces an aggregation pheromone that attracts both sexes and can modify prey responses, including reducing leaf consumption and larval abundance (Hermann and Thaler 2014, Aflitto and Thaler 2020). There are several hypotheses for why this pheromone may affect the community of herbivores or natural enemies in the system. Because this cue

aggregates conspecifics, it may intensify intraspecific competition among naturally occurring or augmented *P. maculiventris* when prey are limited or alter intraguild interactions. *P. maculiventris* is known to act as an intraguild predator on the juveniles of other soft-bodied predators like lady beetles (Hough-Goldstein et al. 1996, Mallampalli et al. 2002). Conversely, herbivores may interpret the same cue as evidence of increased predation risk and adjust their behavior by reducing feeding, avoiding treated areas, or altering oviposition to lower the likelihood of predator encounter (Werner and Peacor 2003, Hermann and Thaler 2014). The synthetic pheromone blend consists of  $\alpha$ -terpineol, terpinen-4-ol, benzyl alcohol, linalool, and (*E*)-2-hexenal (Aldrich et al. 1984). Two components, linalool and (*E*)-2-hexenal, are also green leaf volatiles released by potato plants following herbivore injury, and herbivore-induced plant volatiles (HIPVs) are broadly used by insects across taxa as indicators of herbivore activity, plant damage, and changes in the local risk landscape (Heil 2014). This chemical overlap between predator cues and plant volatiles suggests potential for community-level, non-target effects when the synthetic blend is deployed in the field.

In this study, we evaluate how a synthetic blend mimicking *P. maculiventris* aggregation pheromone influences the target pest the Colorado potato beetle and the associated arthropod communities as well as enemy-pest interactions in potato fields. We hypothesized that the deployment of the predator pheromone would reduce Colorado potato beetle abundance, alter herbivore and predator communities, increase predation on Colorado potato beetle eggs, and influence herbivore and predator feeding behavior ~~to herbivores and predators may alter their feeding behavior in response to~~ *Podisus*-specific cues, either directly or indirectly through pheromone-induced changes in the abundance or ~~feeding~~ behavior of naturally occurring predators. ~~4)~~ We used field experiments to test the effects of the synthetic predator pheromone

on herbivore and natural enemy abundance and diversity using weekly D-vac sampling and visual counts. 2) We tested the effects of pheromone on predation rates by deploying sentinel beetle egg clutches across the potato growing season. 3) We conducted complementary greenhouse assays to test whether the pheromone affected predation rates of beetle eggs by dominant field predators. Understanding these dynamics is crucial for assessing the non-target and community-level impacts of deploying predator pheromones as a novel biocontrol tool in agroecosystems.

## Methods

### 1. Field experiments

#### *Field set-up*

On May 3, 2024, *Solanum tuberosum* (cv. Yukon Gold) potatoes were planted at Cornell University's Homer C. Thompson Vegetable Research Farm in Freeville, NY (42.517895, -76.332011). The variety Yukon Gold was selected due to its extensive use in prior research in this system (Hermann and Thaler, 2014; Aflitto and Thaler, 2020). Before planting, the field was fertilized with Triple 13 NPK at a rate of 1289 kg/ha. The experimental layout consisted of 36 single rows of potatoes, each row measuring 30 meters in length. Potato seeds were planted 0.3 m apart within rows, with a row spacing of 0.9 m. The field was organized into three blocks (A, B, and C), each containing 12 parallel rows. Blocks were separated by approximately four meters of bare soil and were oriented from right (A) to center (B) to left (C) of the site.

#### *Predator pheromone release*

The synthetic predator pheromone used in this study mimics the male aggregation pheromone of *P. maculiventris*, which is naturally released in spring to attract females. The dispenser

containing the predator pheromone was made from a modified opaque brown 2 mL Eppendorf tube with three 0.55 mm holes in the top. Each dispenser was filled with 594.1 µL of an in-house blend of a 594.1 µL of a synthetic male *P. maculiventris* dorsal abdominal gland pheromone mixture (354.0 µL (E)-2-hexenal, 216.0 µL alpha terpineol, 2.4 µL linalool, 2.0 µL terpinen-4-ol, and 19.7 µL benzyl alcohol) (Aldrich et al., 1984) and affixed to a highlighter green marking flag 20 cm high. Control plots received an empty tube attached to a marking flag. Pheromone dispensers have been shown to exert non-consumptive effects within approximately one meter of their placement (Martinez et al. 2025; Aflitto and Thaler 2020). To account for this, plots were designed to be two meters long and included two pheromone dispensers per plot. The dispensers emitted the pheromone at an average rate of 0.23 mg/hr (Aflitto and Thaler 2020). This release rate was selected based on dispenser characterization described by Aflitto and Thaler (2020). Although individual *P. maculiventris* have been reported to produce up to approximately 1 mg of aggregation pheromone (Aldrich et al. 1984), natural pheromone exposure experienced by Colorado potato beetles in the field likely varies with predator density, spatial distribution, and environmental conditions. Consequently, our dispenser was intended to provide a biologically relevant pheromone stimulus rather than to replicate natural field concentrations exactly. ~~Studies have found that a single male *P. maculiventris* can release up to a maximum of 1 mg of the pheromone at a time, indicating that this blend and release rate mimics what the Colorado potato beetle could potentially encounter in the field (Aflitto & Thaler, 2020; Aldrich et al., 1984).~~

Pheromone treatments were deployed on May 30, 2024 to coincide with the emergence of overwintering Colorado potato beetle adults. Within each row, plots consisted of six plants spaced 30 cm apart in a single row, with two pheromone flags placed adjacent to the second and

fourth plants. A five-meter buffer of untreated potato plants was maintained between plots. This spacing was selected based on previous field experiments in our lab indicating no effect on beetle abundance or feeding beyond approximately one meter from the pheromone dispensers and was considered sufficient to minimize potential pheromone movement among adjacent plots (Aflitto & Thaler, 2020; Martinez et. al. 2025). To further reduce side-by-side overlap of treatments, plots were staggered between rows.~~A five-meter buffer of untreated potato plants was maintained between plots. To prevent side-by-side overlap of treatments, plots were staggered between rows.~~

Each row contained two pheromone-treated plots and two control plots. Pheromone-treated plots received two open-dispersion volatile dispensers, while control plots received two empty dispensers. Each plot type included two distinct sampling methods, as described below, and treatments were randomized across all rows for a total of 144 plots. Dispensers were left in the field throughout the measurement period and were replaced every 1.5 weeks.

### ***Field sampling***

We measured the effect of predator pheromone on the abundance of potato herbivores and natural enemies using two sampling methods. Starting June 4, 2024, and continuing weekly for seven weeks, we conducted visual counts of ~~all~~ Colorado potato beetle predators, the Colorado potato beetle (eggs, larvae, and adults), and other potato pests present on the six plants within each of the 72 plots (half control, half with pheromone). Colorado potato beetle predators were identified based on published literature describing established natural enemies of CPB (Weber et al., 2022). ~~Insects were identified visually in the field and were not collected during visual surveys.~~ For each plant, the observers positioned themselves approximately 15–30 cm from the plant material and conducted a systematic inspection. Observers gently manipulated the stems

and foliage to examine both the upper and lower surfaces of all leaves, including those near the base of the plant, to ensure full coverage of the plant canopy. To minimize potential observer bias, observers were rotated among plots during each sampling period so that no observer consistently sampled the same plots. In addition, all observers had prior entomological experience and received additional training before sampling on the identification of arthropod taxa commonly encountered in potato fields. An identification guide was available during each sampling event, and a trained entomologist was available to verify uncertain field identifications. ~~Insects were identified visually in the field and were not collected during visual surveys.~~

In addition to weekly visual estimates, we used a handheld D-vac device (Model 122 Rincon-Vitova Insectaries, Ventura CA) on the other 72 plots (half control, half with pheromone). Each plant was vacuumed for 5–10 seconds. Samples from all six plants were placed into a collection bag labeled with block, plot ID, row, date of collection and placed in the freezer.

### ***Insect Identification***

All observed organisms were identified to family, with taxa of particular importance for Colorado potato beetle predation and potato pest status identified to finer taxonomic resolution when possible. Focal herbivores included the potato leafhopper, *Empoasca fabae* (Harris) (Hemiptera: Cicadellidae); the Colorado potato beetle, *Leptinotarsa decemlineata* (Say) (Coleoptera: Chrysomelidae); aphids (Hemiptera: Aphididae); the tarnished plant bug, *Lygus lineolaris* (Palisot de Beauvois) (Hemiptera: Miridae); and flea beetles (Coleoptera: Chrysomelidae: Galerucinae) (Weber et al. 2022). Predators were identified to family and included lady beetles (Coleoptera: Coccinellidae), minute pirate bugs (Hemiptera: Anthocoridae), spiders (Araneae: Araneidae), damsel bugs (Hemiptera: Nabidae), and lacewings (Neuroptera: Chrysopidae, Hemerobiidae). Species-level identification was conducted for key

predators of the Colorado potato beetle, including *Lebia grandis* (Coleoptera: Carabidae) (Hentz) and *Podisus maculiventris* (Hemiptera: Pentatomidae) (Say). Lady beetles were further identified ~~to as~~ *Coleomegilla maculata* (Coleoptera: Coccinellidae) (De Geer), a native, naturally occurring predator of Colorado potato beetle, whereas other coccinellids were retained at the family level for community analysis. Herbivore and predator taxa representing less than 1% of total observations were not identified further, as these uncommon taxa were unlikely to play a substantial role in pest suppression or plant damage within this system (Chang and Snyder 2004). Although excluding rare taxa may slightly underestimate richness and diversity, these taxa occurred at very low frequencies and were not expected to substantially influence community-level patterns or conclusions.

### ***Egg predation rates in the field***

To quantify Colorado potato beetle egg predation rates in the field over the potato growing season, we deployed sentinel beetle eggs weekly from June 7 to July 19, 2024. We prepared sentinel eggs using three-week-old potted potato plants grown in the greenhouse and fresh egg clutches (20 to 40 eggs per clutch) collected from a lab colony of the Colorado potato beetle. We used a paired design where an egg clutch was glued to the lower side of a leaf (the “predator-exposed” clutch) and another clutch was glued to the lower side of a nearby leaf of the same plant enclosed in a mesh bag (the “control” clutch). Each week, we randomly selected five to seven control and pheromone-treated plots across our study site and deployed one plant with sentinel eggs in each plot for three days (no repeated deployments in the same locations over the season). Ten to fourteen plants with sentinel eggs were deployed in the field for three days each week (giving a total sample size of 84 over the season), with half of the plants in the control plots and half of the plants in the pheromone-treated plots (the plants with sentinel eggs were placed between two field potato

~~plants).~~ The number of lost eggs in each clutch was determined as the difference between the initial number of eggs and the number of remaining intact eggs (partially damaged and shrunk eggs were counted as lost eggs as they were not able to hatch). To account for non-predation egg mortality, background egg loss from the paired control clutch was scaled to the size of the predator-exposed clutch and subtracted from observed egg loss to estimate predation. The average non-predation mortality was low and similar in the control and pheromone-treated plots (mean  $\pm$  SE; control plot:  $0.77 \pm 0.23$  eggs per three days; pheromone-treated plot:  $0.75 \pm 0.27$  eggs per three days). Observations with clutches (either control or predator-exposed) that were missing or hatched were excluded, giving a final sample size of 50 (30 in the control plots vs. 20 in the pheromone-treated plots) in the analysis.; ~~clutches that were missing or hatched were excluded.~~

## 2. Lab experiments

### *Predation assays*

Greenhouse feeding assays were conducted using the dominant Colorado potato beetle egg predators that were also abundant in the field, focusing on the species that could be reliably collected or maintained under laboratory conditions. These included *Coleomegilla maculata* adults and fourth-instar larvae, *Lebia grandis* adults, *Chrysoperla rufilabris* third-instar larvae, and *P. maculiventris* adults. The assays took place in the greenhouse facility at Cornell University in Ithaca, NY. Minute pirate bug *Orius insidiosus* was also assessed through this study due to its high abundance in the field, however there was no clear consumption evidence of eggs, and no further analysis was performed. *Coleomegilla maculata* adults, *Lebia grandis* and *P. maculiventris* adults were field collected from experimental plots located in Freeville, NY in August 2024. *Chrysoperla rufilabris* and *Orius insidiosus* were purchased from ARBICO Organics, Oro Valley, AZ. Additional adult *Coleomegilla maculata* ~~adults~~ and all fourth-instar

253 *Coleomegilla maculata* larvae were taken from our lab-colony reared in a growth chamber  
254 maintained on a 16:8 L:D dark cycle and fed on pea aphids.

255 For the greenhouse assays, predators were randomly selected from the lab colony once predators  
256 started reproducing or assays were done following field collection. Prior to the experiment,  
257 predators were isolated individually without food for 24 hours to standardize hunger levels.

258 Each replicate contained a single potato plant with a set number of *Leptinotarsa decemlineata*  
259 eggs affixed to the abaxial side of a single, mid-height leaf blade. The beetle eggs were collected  
260 from lab-maintained colonies in a growth chamber on a 16:8 L:D dark cycle. Temperature was  
261 maintained at approximately 28.2°C, ~~and a relative humidity of 21.4%.~~ All eggs were collected  
262 within 24 hours before the start of the assays. Plants were spaced equally in the greenhouse,  
263 about 0.5 meters apart. Individual predators were introduced to the plants randomly assigned to  
264 either the control or the treatment group. Plants in the control group were provided rubber septas  
265 charged with 250 microliters of hexane solution and soaked overnight. Plants in the treatment  
266 group had rubber septas charged with 250 microliters of 1% pheromone solution (see above)  
267 diluted with hexane, attached to a leaf around 10-15 cm across from the egg-affixed leaf. All  
268 plants were covered by perforated bread bags and secured at the base of the plant with a rubber  
269 band. Replicates were evenly divided between control and pheromone treatments. Due to space  
270 constraints, the experiments were carried out over sequential trials, each containing an equal  
271 number of control and pheromone replicates. Total eggs consumed were counted after 24 or 48  
272 hours. Predators that died or pupated during the assay were omitted from analysis.

273 The predation assays were done with *C. maculata* adults (n = 128; 63 control, 65 pheromone), *C.*  
274 *maculata* fourth-instar larvae (n = 27; 15 control and 12 pheromone), *L. grandis* adults (n = 32;  
275 16 control and 16 pheromone), *C. rufilabris* third-instar larvae (n = 46; 23 control and 23

pheromone), and *P. maculiventris* adults (n = 62; 31 control and 31 pheromone; this assay was conducted for 48 hours).

### 3. Data analysis

#### *Abundance of the Colorado potato beetle*

We analyzed the abundance of the Colorado potato beetle (*L. decemlineata*) separately for the following biological and statistical reasons: (1) It is the focal pest species in this system. (2) It has known behavioral responses to predator-derived cues (Aflitto & Thaler, 2020; Helms et al., 2019). (3) It was not found in D-vac samples. (4) Its statistical distribution differed from that of other herbivores in the system. We fit a generalized linear mixed-effects model with beetle larval abundance as the response variable, pheromone treatment as the fixed effect, ~~and~~ with week and plot nested within the row and block as the random effects.

#### *Abundance, richness, and diversity of arthropod communities*

To evaluate how pheromone treatment influenced herbivore and predator community structure, we calculated total abundance, mean abundance per taxonomic group, richness, and Shannon diversity for each field sample. Herbivore and predator datasets were first filtered to include only non-predatory taxa or predatory taxa only, and total counts were summed within each combination of block, row, plot, treatment, sampling method, week, and taxonomic group. Community metrics were calculated at the level of taxonomic groups (family-level units), including taxonomic richness (number of families detected), mean abundance per family, and Shannon diversity using the “vegan” package (Oksanen, 2024). Although community metrics were calculated at the family level, which may mask species-specific responses within some groups, potato arthropod communities are typically dominated by a relatively small number of

herbivore and predator taxa. Consequently, we expect family-level metrics to capture the dominant community responses to pheromone deployment, while recognizing that species-level responses within families may have gone undetected.

Abundance was log-transformed to meet model assumptions and analyzed with a linear mixed-effects model including treatment as the fixed effect and week and plot nested within block as the random effects. Richness was modeled using a generalized linear mixed model with a Poisson error distribution, with treatment included as the fixed effect and week and plot nested within block as the random effects. This model was fitted via the `glmmTMB()` function in the R “glmmTMB” package (Brooks et al. 2017). The effects of treatment on Shannon diversity were evaluated using a linear mixed-effects model with treatment as the fixed effect and week and plot nested within block as the random effects. We checked model assumptions using the simulated scaled residuals generated from the `simulateResiduals()` function in the R “DHARMA” package (Hartig 2024).

### *Abundance of selected predators and herbivores*

To evaluate the effects of pheromone treatment on herbivorous and predator arthropod abundance across taxonomic groups, we constructed generalized linear mixed models using the `lme4` package (Bates et al. 2015). Herbivore and predator datasets were filtered to include taxa that were consistently detected across sampling events. The selected herbivores included *Cicadellidae* (potato leaf hopper), *Galerucinae* (flea beetles), *Miridae* (tarnished plant bug), and *Aphididae* (Aphids); the selected predators included *P. maculiventris*, Neuroptera (lacewings), *Coccinellidae* (lady beetles), *Nabidae* (damselflies), *Araneidae* (spiders), and *Anthrocoridae* (minute pirate bugs). Abundance for each taxonomic group was calculated as the mean number

of individuals collected within each combination of block, plot, row, treatment, sampling method, and week.

We modeled herbivore abundance as a function of treatment, taxonomic group, and their interaction using a negative binomial GLMM. The full model included the interaction between treatment and taxonomic group as fixed effects, with plot nested within block and week included as random effects. A negative binomial error distribution was used to account for overdispersion in count data. Post hoc comparisons among treatment levels within each taxonomic group were performed using pairwise contrasts of estimated marginal means (EMMs) with the emmeans package (Lenth, 2025<sup>4</sup>).

### ***Egg predation in the field***

To compare the egg predation rates between the control and the pheromone-treated plots, we fit a generalized linear model with the number of predated eggs as the response, and plot treatment type and week as the predictors. We used a negative binomial distribution with a log link function to account for data overdispersion. The model was fitted via the `glmmTMB()` function in the R “glmmTMB” package (Brooks et al. 2017). We checked model assumptions using the simulated scaled residuals generated from the `simulateResiduals()` function in the R “DHARMa” package (Hartig 2024). We used the Wald chi-square test to assess predictor significance using the `Anova()` function (type II sums of squares) in the R “car” package (Fox & Weisberg 2018). The analysis was performed in R version 4.3.3 (R Core Team 2024).

### ***Egg predation in the greenhouse***

We analyzed the probability of egg consumption by four predator species (*P. maculiventris*, *C. maculata*, *L. grandis*, and *C. rufilabris*) using a generalized linear mixed model fitted with the

glmmTMB package (Brooks et al. 2017). The response variable was the number of eggs consumed relative to the number available at the start of the assay, modeled as a binomial response. Fixed effects included treatment, predator life stage/species, their interaction, and assay length as a covariate. Random effects included trial identity since experiments were conducted sequentially, and plant identity. -The species *C. maculata* was separated to analyze the treatment effect on egg consumption in larvae and adult stages. *Coleomegilla maculata* was selected because it represented the highest proportion of lady beetles observed in the field, while *L. grandis* was included due to its role as an important specialist predator of the Colorado potato beetles. All analysis was conducted in R version 4.4.0 (R Core Team 2025).

## Results

### *Herbivore abundance and community structure*

While the abundance of the target herbivore, the Colorado potato beetle, was reduced by the pheromone, none of the other herbivores were affected. On average, the number of Colorado potato beetle larvae was significantly lower in the pheromone-treated plots compared to the control plots ( $Z = -2.51$ ;  $p = 0.01$ ; Fig. 1). The estimated mean abundance of Colorado potato beetle larvae was 32% lower in pheromone-treated plots than in control plots (5.53 vs. 8.08 larvae per plot;  $z = -2.51$ ,  $P = 0.01$ ; Fig. 1).

The presence of the synthetic predator pheromone did not significantly alter herbivore abundance, richness, or diversity of the rest of the herbivore community over the seven-week sampling period (Appendix S1; Fig. S1; Table S1). The herbivore community was dominated by potato leafhoppers, which comprised approximately 79% of all herbivores observed, followed by the Colorado potato beetles (6%), aphids (5%), tarnished plant bugs (5%), and flea beetles (4%). Together, these taxa accounted for approximately 99% of the herbivore community.

Aphids, potato leafhoppers, flea beetles, and tarnished plant bugs did not differ significantly in abundance between the control and pheromone treatments (Fig. 2; Table 1).

### ***Predator community structure***

Exposure to the synthetic predator pheromone did not significantly alter predator abundance, richness, or diversity over the seven-week sampling period (Appendix S1; Fig. S2; Table S2).

The predator community was dominated by lady beetles, which comprised approximately 48% of all predators observed, followed by minute pirate bugs (22%), spiders (12%), damsel bugs (6%), spined soldier bugs (5%), and lacewings (4%). Together, these taxa accounted for approximately 97% of the predator community.

The most abundant predators in the potato field—minute pirate bugs, spiders, lady beetles, damsel bugs, lacewings, and spined soldier bugs, did not differ significantly in abundance between the control and pheromone treatments (Fig. 3; Table 2).

### ***Predation rates of Colorado potato beetle eggs in the field***

Overall egg predation rates over a two-month period were higher in the pheromone-treated plots (mean [95% CI]: 9.8 [2.7–15.9] eggs per three days) compared to the control plots (mean [95% CI]: 1.4 [0.4–2.4] eggs per three days) ( $\chi^2 = 11.7$ ;  $df = 1$ ;  $p < 0.001$ ; Fig. 4).

~~Overall egg predation rates over a two-month period were higher in the pheromone-treated plots compared to the control plots ( $\chi^2 = 11.7$ ;  $df = 1$ ;  $p < 0.001$ ; Fig. 4).~~

### ***Predation of Colorado potato beetle eggs in greenhouse assays***

There is no significant effect of pheromone treatment on the proportion of eggs consumed by adult *C. maculata* (81.9% vs. 82.9%;  $Z = 0.09$ ;  $p = 0.92$ ; Fig. 5), *C. maculata* larvae (85.5% vs.

386 68.3%;  $Z = 0.83$ ;  $p = 0.41$ ); *C. rufilabris* (20.7% vs. 13.9%;  $Z = 0.52$ ;  $p = 0.60$ ; Fig. 5), or *L.*  
 387 *grandis* (31.1% vs. 66.1%;  $Z = -1.30$ ;  $p = 0.19$ ; Fig. 5).

388 There is a significant effect of pheromone treatment on the proportion of eggs consumed by *P.*  
 389 *maculiventris* ( $Z = -3.69$ ;  $p < 0.001$ ; Fig. 5) with estimated egg consumption increasing from  
 390 1.3% in the control treatment to 29.2% in the pheromone treatment. ~~where stink bug adults~~  
 391 ~~consumed higher proportions of eggs when the pheromone was present.~~

## 392 Discussion

393 While the abundance of the target herbivore (the Colorado potato beetle) was reduced by  
 394 exposure to the synthetic predator pheromone, the abundance, species richness, and diversity of  
 395 herbivores and natural enemies in the field were not altered throughout the season (Appendix S1;  
 396 Fig. S3-S4). Despite these neutral community responses, predation on beetle eggs was  
 397 significantly higher in pheromone-treated plots, suggesting that the pheromone may have  
 398 influenced predator foraging behavior rather than predator abundance. ~~However, predator~~  
 399 ~~behavior was not directly measured in this study.~~ ~~eue elicited changes in predator behavior rather~~  
 400 ~~than changes in predator abundance.~~ Greenhouse assays further revealed that *P. maculiventris*  
 401 increased its feeding rate in the presence of the pheromone, demonstrating that the compound  
 402 can enhance consumptive effects for at least one key predator species. In contrast, *C. maculata*  
 403 lady beetle larvae and adults-, lacewing larvae, and *L. grandis* showed no detectable response.  
 404 Collectively, although predators varied in their feeding responses to the synthetic pheromone, the  
 405 net outcome positively favored future use of the pheromone in biological control of the Colorado  
 406 potato beetle.

The lack of community-wide responses by herbivores may reflect differences in how taxa perceive or respond to predator-associated volatiles. Most herbivores in this system, aside from the Colorado potato beetle, are not known to be preyed upon by *P. maculiventris* and therefore may have little incentive to respond directly to its cues. Although previous work demonstrates that predator pheromones can alter plant physiology and quality, both in potatoes exposed to the *P. maculiventris* pheromone (Aflitto & Thaler, 2020; Martinez et al., 2025) and in other systems where plants respond to predator-derived cues (Helms et al., 2017), the herbivores in our study did not show detectable changes in a way that suggested response to plant quality. Consistent with this, the Colorado potato beetle responded to predator-pheromone-induced changes in plant quality in ~~earlier experiments~~, previous greenhouse experiments (Martinez et al., 2025), but we found no evidence that other herbivores reacted similarly in our field surveys. Many chewing herbivores adjust behavior rather than population numbers in response to predator cues (Kaplan et al., 2014), though such subtle shifts may have gone undetected as we did not measure behavior. Additionally, herbivores such as leafhoppers and aphids rely heavily on plant nutritional quality and phloem chemistry, traits that may not be strongly influenced by exposure to the predator pheromone (Hewer et al., 2010). Chewing herbivores like flea beetles, despite sharing a feeding mode with Colorado potato beetle larvae, may also differ in how they perceive and respond to predator-associated volatiles (Li et al., 2024).

Aggregation pheromones have been shown to attract adult *P. maculiventris* in field settings, yet such recruitment was not detected in our experiment (Kelly et. al 2014). Predators may vary widely in their responses to the pheromone. Notably, *C. maculata* has been shown to respond to a component of the *P. maculiventris* pheromone under field conditions (Zhu et al. 1999; Kaplan 2012), suggesting capacity for attraction in at least some members of the predator

guild. There is evidence that predators can be attracted by HIPV<sup>2</sup>s in field settings; for example, methyl salicylate reliably pulls in diverse predator guilds, showing that single semiochemicals can serve as attractive cues (James, 2005; Zhang et al., 2006; Zhu & Park, 2005). Additionally, our D-vac and visual sampling methods quantified predator abundance rather than predator activity and may not have detected localized changes in predator behavior within plots. However, predator abundance in our fields remained unchanged across treatments, and predators did not appear to increase their presence in plots with higher Colorado potato beetle larval abundance. This indicates that the pheromone was neither a deterrent nor an attractive cue at the plot scale, nor were any patches of larvae serving as a prey hotspot. ~~However, P~~predation rates on Colorado potato beetle eggs changed in the field, prompting us to examine predator feeding behavior in greenhouse assays, where we did observe species-specific responses to the pheromone. *Podisus maculiventris* increased feeding, which may be due to the species' established use of aggregation pheromones in mate attraction and prey sharing (Sant'Ana et al., 1997). Alternatively, increased egg consumption may reflect a behavioral response to aggregation cues that elevate local predator density, as higher conspecific densities commonly alter per-capita feeding rates through interference or competitive interactions in biological control systems (DeLong & Vasseur, 2011). However, this mechanism was not directly tested in our study and warrants future investigation. Although not statistically significant, *L. grandis* exhibited a moderate increase in egg consumption under pheromone exposure, suggesting potential responsiveness among some beetle predators. Not all predators exhibited behavioral responses to the pheromone treatment. *C. maculata* adults did not respond to the predator pheromone. While there is evidence of adult response to an individual component of the pheromone (Zhu et al. 1999; Kaplan 2012) this sensitivity may not translate into attraction or

altered feeding behavior in response to a more complex pheromone blend. Similarly, *C. maculata* larvae and lacewings showed no significant differences in feeding rates, suggesting that the pheromone does not universally enhance predation across the natural enemy guild. This lack of response is noteworthy given that these larval predators could be at risk of intraguild predation by adult stink bugs (Abram et al., 2014; Mallampalli et al., 2002). Predator-derived cues have been shown to trigger behavioral and physiological responses in insect prey, though whether such effects consistently scale up to influence population abundance or community structure remains poorly resolved (Hermann & Landis, 2017). Such asymmetries highlight the importance of considering functional traits and ecological roles of the larger community when evaluating the potential of semiochemicals as biocontrol tools.

From an applied perspective, our findings demonstrate that predator-derived pheromones can enhance biological control. By increasing predation on Colorado potato beetle eggs, the pheromone has the potential to improve the biocontrol efficiency of key predators, particularly given that Colorado potato beetle feeding damage can reduce potato yield when defoliation exceeds the plants compensatory capacity translates directly into yield loss (Hare, 1990; Stieha and Poveda, 2015). The low numbers of *P. maculiventris* observed in the field suggest that the increased predation in pheromone-treated plots likely resulted from subtle shifts in predator behavior rather than from increases in predator abundance. This is notable because *P. maculiventris* is difficult to retain in cropping systems and often requires multiple augmentative releases (Aldrich & Cantelo, 1999; Biever & Chauvin, 1992; Tipping et al., 1999); even a small increase in conspecific attraction, combined with higher per-capita feeding rates, could meaningfully improve its efficacy. Importantly, this enhancement occurred without detectable reductions in the abundance of either beneficial insects or non-target herbivores, indicating that

pheromone deployment is unlikely to generate the unintended impacts commonly associated with broad-spectrum insecticides. Although these findings demonstrate the potential of predator pheromones for enhancing biological control, our conclusions are based on a single growing season at one location in a potato production system using a single pheromone release rate. Future studies evaluating multiple environments, cropping systems, years, and pheromone release rates will be important to determine the generality and optimize the application of this approach. Collectively, these outcomes support the integration of predator pheromones into push–pull or attract-and-stimulate strategies to strengthen pest suppression in potato systems.

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636 **Figure 1:** Mean ( $\pm$  SE) number of Colorado potato beetle larvae per plot in control and  
637 pheromone-treated plots averaged across the sampling period. Raw observed means were  $8.51 \pm$   
638  $0.89$  larvae per plot in the control treatment ( $n = 177$  observations) and  $6.73 \pm 0.82$  larvae per

plot in the pheromone treatment ( $n = 159$  observations). Error bars represent  $\pm 1$  SE. Asterisk indicates a significant treatment effect ( $p < 0.05$ ). Number of the Colorado potato beetle larvae in the control and the pheromone-treated plots averaged over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ( $p < 0.05$ ).

**Alt text 1:** Bar plot showing number of Colorado potato beetle larvae in the control (yellow bar) and the predator pheromone (purple bar) plots. There are less beetles in the plots with the predator pheromone.

**Figure 2.** Average herbivore number per plot across two months in control and pheromone-treated plots. Herbivores are shown by family and were the major herbivore families found in the field. Aphididae (aphids), Cicadellidae (potato leafhopper), Galerucinae (flea beetles), and Miridae (tarnished plant bugs). Error bars represent standard errors.

**Alt text 2:** Number of herbivores in each plot (control or pheromone). Listed by different families. From right to left – Aphididae, Cicadellidae, Galerucinae, Miridae.

**Figure 3.** Average predator number per plot across two months in control and pheromone treated plots. Predators are shown by family and are the most abundant in the field, Anthocoridae, Araneidae, Neuroptera (Chrysopidae/Hemerobiidae), Coccinellidae, Nabidae, and Pentatomidae~~Neuroptera~~. Error bars represent standard errors.

**Alt text 3:** Number of predators in each plot (control or pheromone). Listed by different families. From right to left – Anthocoridae, Araneidae, Chrysopidae/Hemerobiidae, Coccinellidae, Nabidae, Pentatomidae.

**Figure 4.**

Overall beetle egg predation rates in the control (mean [95% CI]: 1.4 [0.4–2.4]) and the pheromone-treated plots (mean [95% CI]: 9.8 [2.7–15.9]) over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ( $p < 0.05$ ).

~~Overall beetle egg predation rates in the control and the pheromone-treated plots over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ( $p < 0.05$ ).~~

**Alt text 4:** Predation rates of predators found in potato field. Rates represent eggs consumed every three days in control or pheromone treated plots.

**Figure 5.** Percent of the Colorado potato beetle eggs consumed by five predator species/life stages when exposed to predator pheromone or control volatiles in greenhouse assays. Panels from left to right show *Coleomegilla maculata* adults, *Chrysoperla rufilabris*, *Lebia grandis*, *Podisus maculiventris*, and *Coleomegilla maculata* larvae. Error bars represent standard errors. Asterisks denote significant differences between treatment levels ( $p < 0.05$ ).

**Alt text 5:** Percent of Colorado potato beetle eggs consumed by predators in greenhouse assays. From right to left – *Coleomegilla maculata* adults, *Coleomegilla maculata* larvae, *Lebia grandis*, *Chrysoperla rufilabris*, *Podisus maculiventris*.

| Taxonomic group     | Control /<br>Pheromone ratio | SE   | Z     | p-value |
|---------------------|------------------------------|------|-------|---------|
| <i>Aphididae</i>    | 1.10                         | 0.09 | 1.09  | 0.28    |
| <i>Cicadellidae</i> | 1.07                         | 0.07 | 1.04  | 0.29    |
| <i>Galerucinae</i>  | 0.93                         | 0.09 | -0.72 | 0.47    |
| <i>Miridae</i>      | 1.10                         | 0.09 | 0.14  | 0.89    |

**Table 1.** Control /pheromone abundance ratios ( $\pm$  SE) for dominant herbivore taxa. Ratios were estimated from generalized linear mixed models; values significantly different from 1 would indicate treatment effects. No significant differences were detected ( $p < 0.05$ ).

| Taxonomic Group      | Control /<br>Pheromone ratio | SE   | Z     | p-value |
|----------------------|------------------------------|------|-------|---------|
| <i>Anthocoridae</i>  | 1.11                         | 0.17 | 0.69  | 0.49    |
| <i>Araneidae</i>     | 0.81                         | 0.25 | -0.68 | 0.50    |
| <i>Coccinellidae</i> | 0.99                         | 0.09 | -0.14 | 0.89    |
| <i>Nabidae</i>       | 0.92                         | 0.24 | -0.33 | 0.74    |
| <i>Neuroptera</i>    | 1.12                         | 0.35 | 0.46  | 0.65    |
| <i>Podisus</i>       | 1.11                         | 0.38 | 0.36  | 0.76    |

**Table 2.** Control/pheromone abundance ratios ( $\pm$  SE) for dominant predator taxa. Ratios were estimated from generalized linear mixed models; values significantly different from 1 would indicate treatment effects. No significant differences were detected ( $p < 0.05$ ).

Dear Editors and the Reviewers:

Thank you so much for the opportunity to revise our manuscript. My co-authors and I appreciate the constructive suggestions and advice that you and the reviewers provided. We have revised our manuscript accordingly (see our responses in blue text below). Both our final revised manuscript and the revised manuscript with changes tracked have been uploaded.

### **Specific Comments with line numbers**

Lines 13-14. Which species of common predators?

**Response:** We thank the reviewer for this helpful suggestion. To provide additional clarity while maintaining the brevity required of the abstract, we have revised the text to indicate that the greenhouse assays evaluated five predator taxa rather than using the more general phrase "common predators." The specific predator species and life stages are described in detail in the Methods and Results sections.

**Revision:** Greenhouse feeding assays tested whether five predator taxa altered their consumption of *L. decemlineata* eggs in response to the pheromone.

Lines 74-75. Need to add a few sentences describing the importance of the Colorado potato beetle (see, for example, Alyokhin et al. 2022. Colorado potato beetle. In: Alyokhin, A., S.I. Rondon, and Y. Gao [editors]. Insect Pests of Potato: Global Perspectives on Biology and Management. Second Edition. Academic Press, Oxford, UK. Pp. 29-43).

**Response:** We thank the reviewer for this suggestion. Because the Introduction is primarily focused on predator pheromones rather than the biology of the Colorado potato beetle, we chose not to substantially expand this section. However, to provide additional context, we have added a brief sentence highlighting the agricultural importance of the Colorado potato beetle as a major pest of potato production.

**Revision:** This study focuses on one such system involving the spined soldier bug, *Podisus maculiventris* (Say) (Hemiptera: Pentatomidae), a generalist predator and key natural enemy of the Colorado potato beetle (*Leptinotarsa decemlineata*) (Say) (Coleoptera: Chrysomelidae). Colorado potato beetle is among the most destructive insect pests of potato worldwide because of its capacity for rapid defoliation and widespread insecticide resistance (Alyokhin et al. 2022).

Lines 98-103. I do not think that numbering these sentences is necessary.

**Response:** We thank the reviewer for this suggestion. We agree that numbering the study objectives is unnecessary in this context and have removed the numbering to improve the flow of the text.

Lines 116-117. Four meters of what? Bare soil or potatoes?

**Response:** We thank the reviewer for pointing out this ambiguity. We have clarified that the approximately four-meter separation between blocks consisted of bare soil.

**Revision:** Blocks were separated by approximately four meters of bare soil and were oriented from right (A) to center (B) to left (C) of the site.

Lines 122-125. Was the pheromone blended “in-house” by the authors? Or was it obtained from an external source?

**Response:** We thank the reviewer for this comment. We have clarified that the pheromone was an in-house blend prepared by the authors according to the formulation described by Aldrich et al. (1984).

**Revision:** Each dispenser was filled with 594.1  $\mu$ L of an in-house blend of a synthetic male *P. maculiventris* dorsal abdominal gland pheromone mixture (354.0  $\mu$ L (E)-2-hexenal, 216.0  $\mu$ L alpha terpineol, 2.4  $\mu$ L linalool, 2.0  $\mu$ L terpinen-4-ol, and 19.7  $\mu$ L benzyl alcohol) (Aldrich et al., 1984)...

Lines 171-173. Was that the only species present? Or were the rest of the species pooled?

**Response:** We thank the reviewer for this comment. Other lady beetle species were occasionally observed in the field; however, *Coleomegilla maculata* was the dominant native coccinellid predator in our study system and the only lady beetle species identified to the species level. Other coccinellids were retained at the family level (Coccinellidae) for community analyses. We have clarified this in the Methods.

**Revision:** Lady beetles were further identified to *Coleomegilla maculata* (Coleoptera: Coccinellidae) (De Greer), a native, naturally occurring predator of Colorado potato beetle, whereas other coccinellids were retained at the family level for community analysis

Lines 201-206. It is not clear whether *Coleomegilla maculata* adults were collected in the field, or obtained from the colony, or both.

**Response:** We thank the reviewer for identifying this ambiguity. We have revised the text to clarify that adult *Coleomegilla maculata* used in this study were obtained from both field collections and our laboratory colony, whereas fourth-instar larvae were obtained exclusively from the laboratory colony.

**Revision:** *Coleomegilla maculata* adults, *Lebia grandis* and *P. maculiventris* adults were field collected from experimental plots located in Freeville, NY in August 2024. *Chrysoperla*

*rufilabris* and *Orius insidiosus* were purchased from ARBICO Organics, Oro Valley, AZ. Additional adult *Coleomegilla maculata* and all fourth-instar *Coleomegilla maculata* larvae were taken from our lab-colony reared in a growth chamber maintained on a 16:8 L:D dark cycle and fed on pea aphids.

Lines 230-296. I think that it is better to split this section and place descriptions of statistical analyses after the descriptions of experimental designs.

**Response:** We thank the reviewer for this suggestion. We chose to retain a dedicated Data Analysis section because it provides a centralized description of the statistical methods used across multiple experiments, avoids repetition of similar analytical approaches, and improves the overall readability of the Methods. We believe this organization allows readers to distinguish the experimental procedures from the statistical analyses while maintaining a clear and concise presentation.

Line 353. Need to specify which earlier experiments.

**Response:** We thank the reviewer for this comment. We agree that the reference to "earlier experiments" was ambiguous. We have revised the text to explicitly refer to our greenhouse plant induction experiments (Martinez et al. 2025), which demonstrated that Colorado potato beetle responded to predator-pheromone-induced changes in plant quality.

**Revision:** Consistent with this, the Colorado potato beetle responded to predator-pheromone-induced changes in plant quality in previous greenhouse experiments (Martinez et al., 2025), but we found no evidence that other herbivores reacted similarly in our field surveys

Lines 398-400. I am not sure that the word "directly" should be used here. Potatoes can tolerate considerable defoliation without suffering reduction in yield (see, for example, Stieha and Poveda, Ann Appl Biol 166 [2015] 208–217).

**Response:** We thank the reviewer for this helpful comment. We agree that the term "directly" oversimplifies the relationship between Colorado potato beetle feeding damage and potato yield. We have revised the text to acknowledge that yield loss depends on the extent of feeding damage and the plant's capacity to compensate, while retaining the broader point that reducing Colorado potato beetle injury has the potential to improve crop yield.

**Revision:** By increasing predation on Colorado potato beetle eggs, the pheromone has the potential to improve the biocontrol efficiency of key predators, particularly given that Colorado potato beetle feeding damage can reduce potato yield when defoliation exceeds the plants compensatory capacity (Hare, 1990; Stieha and Poveda, 2015).

## Abstract

### Lines 17–20

The statement that predation rates were "five times higher" should include actual means or percentages to provide biological context.

**Response:** Thank you for pointing this out. For transparency, the actual means are now reflected in the revision.

**Revised:** However, predation rates in pheromone-treated plots (mean  $\pm$  SE:  $9.8 \pm 2.9$  eggs consumed per three days) were five times higher compared to the control ( $1.4 \pm 0.5$  eggs consumed per three days).

### Lines 18–20

The abstract implies that enhanced predation explains beetle suppression. This conclusion is plausible but not directly demonstrated because larval abundance and egg predation were measured independently.

**Response:** We thank the reviewer for this observation. We agree that our original wording implied a direct causal link between enhanced predation and reduced beetle abundance, whereas these responses were measured in separate experiments. We have revised the abstract to clarify that enhanced predation is a potential mechanism supported by the greenhouse assays rather than a directly demonstrated explanation for the field reduction in *L. decemlineata* populations.

**Revised:** Greenhouse assays showed increased egg consumption by *P. maculiventris*. These findings suggest that the synthetic predator pheromone may enhance the foraging behavior of *P. maculiventris*, representing a potential mechanism underlying the observed reduction in *L. decemlineata* populations and a promising tool for potato pest management with minimal non-target impacts on herbivore and predator communities.

## Introduction

### Lines 34–40

Consider adding another example of successful aggregation pheromone use in open-field agriculture beyond cucumber beetles and thrips.

**Response:** We thank the reviewer for this helpful suggestion. We have expanded the Introduction to include an additional example of successful aggregation pheromone use in agriculture. Specifically, we now highlight the synthetic aggregation pheromone of the boll weevil (*Anthonomus grandis*), which has been widely used for monitoring and mass trapping in cotton production (Leonhardt et al. 1988).

**Revision:** A well-known example is the synthetic aggregation pheromone of the boll weevil (*Anthonomus grandis*), which has been widely used for monitoring and mass trapping in cotton

production (Leonhardt et al. 1988). Similarly, aggregation pheromones have been successfully applied in other agricultural systems.

**Revision:**

**Lines 58–65**

The discussion of non-consumptive effects would benefit from mentioning that predator cues can also induce physiological responses in plants, which may indirectly affect herbivores.

**Response:** We thank the reviewer for this helpful suggestion. We have expanded the Introduction to acknowledge that predator-associated cues can also induce physiological responses in plants, which may indirectly influence herbivore behavior and performance.

**Revision:** In addition to directly influencing herbivore behavior, predator-associated cues can induce physiological and defensive responses in plants, which may further reduce herbivore performance through plant-mediated mechanisms.

**Lines 94–103**

The hypotheses are not clearly stated. Consider explicitly listing predictions, For instance: reduced CPB abundance, altered herbivore community composition, altered predator abundance or diversity, increased predation rates.

**Response:** We thank the reviewer for this suggestion. We have revised the final paragraph of the Introduction to explicitly state our study predictions

**Revision:** We hypothesized that the deployment of the predator pheromone would reduce Colorado potato beetle abundance, alter herbivore and predator communities, increase predation on Colorado potato beetle eggs, and influence herbivore and predator feeding behavior to *Podisus*-specific cues, either directly or indirectly through pheromone-induced changes in the abundance or behavior of naturally occurring predators

**Lines 97–98**

The sentence appears incomplete:

"pheromone-induced changes in the abundance or feeding behavior of naturally occurring predators."

Please revise for clarity.

**Response:** We thank the reviewer for pointing out the lack of clarity. The original sentence was incomplete and did not clearly convey the hypothesized indirect mechanism. We have revised the sentence to clarify that we hypothesize the predator pheromone may influence herbivore and predator behavior both through direct responses to *Podisus*-specific cues and indirectly through changes in the abundance or feeding behavior of naturally occurring predators.

**Revised:** We hypothesize .....*Podisus*-specific cues, either directly or indirectly through pheromone-induced changes in the abundance or feeding behavior of naturally occurring predators.

## Methods

### Field Design

#### Lines 128–133

Please provide a stronger justification for the selected release rate and explain how it compares to natural pheromone emissions under field conditions. The authors mentioned something about 1 mg is likely what CPB experience in the field, but that assumption is not entirely valid, and there will likely be multiple female that can simultaneously release the pheromones.

**Response:** We thank the reviewer for this important comment. We agree that natural pheromone exposure experienced by Colorado potato beetles in the field is difficult to estimate and likely varies with predator density, spatial distribution, and environmental conditions. We did not intend to imply that our dispensers replicated natural field concentrations. Rather, the release rate ( $0.23 \text{ mg h}^{-1}$ ) was selected because it was previously characterized for this dispenser system (Aflitto & Thaler 2020) and falls within the range of pheromone production reported for individual *P. maculiventris* adults (Aldrich et al. 1984). We have revised the manuscript to clarify this point and acknowledge that future studies examining multiple release rates will be valuable for optimizing field applications.

**Revision:** This release rate was selected based on dispenser characterization described by Aflitto and Thaler (2020). Although individual *P. maculiventris* have been reported to produce up to approximately 1 mg of aggregation pheromone (Aldrich et al. 1984), natural pheromone exposure experienced by Colorado potato beetles in the field likely varies with predator density, spatial distribution, and environmental conditions. Consequently, our dispenser was intended to provide a biologically relevant pheromone stimulus rather than to replicate natural field concentrations exactly.

#### Lines 137–138

Five-meter buffers may not fully prevent pheromone contamination between plots. Please discuss why this distance was considered sufficient.

**Response:** We thank the reviewer for this comment. We acknowledge that a 5-m buffer cannot guarantee complete separation of pheromone plumes under all environmental conditions. The 5-m buffer was selected based on previous field studies in our lab that detected no measurable behavioral response beyond approximately 1 m from the pheromone dispensers. Therefore, a 5-m buffer was considered sufficient to minimize the potential for pheromone movement among

adjacent plots while maintaining adequate replication within the field site. We have added this clarification to the manuscript.

**Revision:** A five-meter buffer of untreated potato plants was maintained between plots. This spacing was selected based on previous field experiments in our lab indicating no effect on beetle abundance or feeding beyond approximately one meter from the pheromone dispensers and was considered sufficient to minimize potential pheromone movement among adjacent plots (Aflitto & Thaler, 2020; Martinez et. al. 2025). To further reduce side-by-side overlap of treatments, plots were staggered between rows.

## Field Sampling

### Lines 145–158

Were the same observers used throughout the study? If multiple observers participated, explain how observer bias was handled.

**Response:** We thank the reviewer for this comment. Multiple observers participated in sampling throughout the study. To minimize potential observer bias, observers were rotated among plots during each sampling period so that no individual consistently sampled the same plots. We have added this information to the Methods.

**Revised:** To minimize potential observer bias, observers were rotated among plots during each sampling period so that no observer consistently sampled the same plots.

### Lines 147-148

What do the authors mean by all CPB predators? Do you have a knowledge of what species feed on CPB, eggs and larvae? This needs to be better explained here, as such information requires molecular gut content analysis to count a species as predator of CPB. Anyhow, detail information regarding what species were considered CPB predators is needed, to better understand and evaluate this method.

**Response:** We thank the reviewer for pointing out the need for clarification. Our intent was not to infer predation based on field observations alone, nor to identify predators through gut content analysis. Rather, we classified "CPB predators" as arthropod taxa that have been previously documented in the literature to prey on one or more life stages of *Leptinotarsa decemlineata*. We have revised the manuscript to clarify this definition and have added an appropriate reference (Weber et al., Chapter 13, *Biological and Behavioral Control of Potato Insect Pests*) supporting the classification of these taxa.

**Revised:** Colorado potato beetle predators were identified based on published literature describing established natural enemies of CPB (Weber et al., 2022).

Major concern: Carabid beetles are among the most important predatory groups of CPB, and none of the proposed predatory data collection techniques will reliably estimate their density in potato plots. These predators are known to be nocturnal, and hence pitfall traps are the only reliable method for their collection. How do authors justify the techniques used in this experiment? Why did not include pitfalls?

**Response:** We thank the reviewer for this important comment. We agree that pitfall traps are the preferred method for sampling many ground-dwelling carabid beetles, particularly nocturnal species. However, our primary carabid species of interest was *Lebia grandis*, a specialist predator of Colorado potato beetle that differs from most carabids by actively foraging on potato foliage (Weber et al., 2008; Cornell Biological Control Program). Consequently, visual surveys were considered appropriate for detecting foliar-active *L. grandis*. We acknowledge, however, that our daytime visual and D-vac sampling may have underestimated nocturnal carabid activity, including periods when *L. grandis* was inactive on foliage. Some nighttime sampling did occur (3 times) throughout the summer, but few to no predators were visually identified and this was therefore excluded from this manuscript.

#### Lines 153–154

Since insects were not collected, describe how field identifications were validated. How skilled the team were, and how that was factored into data accuracy.

**Response:** We thank the reviewer for this comment. All members of the sampling team had prior entomological experience. To ensure consistent and accurate field identifications, all observers received training before sampling on the identification of the arthropod taxa commonly encountered in potato fields. Observers were provided with an identification guide for reference during sampling, and a trained entomologist was available throughout the study to verify uncertain identifications in the field. We have added this information to the Methods.

**Revision:** All observers had prior entomological experience and received additional training before sampling on the identification of arthropod taxa commonly encountered in potato fields. An identification guide was available during each sampling event, and a trained entomologist was available to verify uncertain field identifications.

## Insect Identification

#### Lines 160–175

Family-level identification may mask treatment effects on particular predator species. Please justify why family-level resolution was sufficient for community analyses.

**Response:** We thank the reviewer for this thoughtful comment. Our objective was to evaluate community-level responses to the predator pheromone while focusing on taxa with the greatest ecological relevance to Colorado potato beetle and potato production. Accordingly, organisms were identified to family level for the community analyses, with finer taxonomic resolution applied to species known to play important roles in this system, including *Podisus maculiventris*, *Lebia grandis*, and *Coleomegilla maculata*. We believe this approach provided sufficient taxonomic resolution to detect treatment effects on the dominant herbivore and predator groups while remaining feasible for repeated field sampling across the growing season.

#### **Lines 173–175**

Excluding taxa representing less than 1% of observations could bias richness and diversity estimates. Please explain how this exclusion may influence results. Given the sight identification in fields, how confident were the team on accurately selecting these taxa?

**Response:** Taxa representing less than 1% of total observations were not identified further because they occurred infrequently and were unlikely to meaningfully influence estimates of abundance or the dominant community patterns examined in this study. We acknowledge that excluding rare taxa may slightly underestimate richness or diversity; however, because these taxa were uncommon and occurred at very low frequencies, we do not expect this decision to have qualitatively affected our conclusions. We have clarified this rationale in the manuscript.

**Revision:** Although excluding rare taxa may slightly underestimate richness and diversity, these taxa occurred at very low frequencies and were not expected to substantially influence community-level patterns or conclusions.

### **Sentinel Egg Predation**

#### **Lines 183–191**

Were sentinel plants randomized among plots each week? Also, authors should clarify whether repeated deployments occurred in the same locations.

**Response:** We randomly selected control and pheromone-treated plots each week to deploy sentinel eggs across the study site, so there were no repeated deployments in the same locations over time. We have clarified this in the methods.

**Revision:** Each week, we randomly selected five to seven control and pheromone-treated plots across our study site and deployed one plant with sentinel eggs in each plot for three days (no repeated deployments in the same locations over the season).

#### **Lines 184–185**

The sample size description is confusing. Please specify: 1) number of plots sampled per week,

2) number of egg clutches per treatment per week, and 3) total number of observations included in analyses.

**Response:** Thanks for pointing this out. Each week, we randomly selected five to seven control and pheromone-treated plots and deployed one plant with sentinel eggs in each plot for three days. In other words, 10 to 14 plots were sampled per week, with five to seven paired control–predator-exposed clutches in each treatment. We excluded some observations with clutches that were missing or hatched, which gives a final sample size of 50 in our analysis. We have clarified this in the methods.

**Revision:** Each week, we randomly selected five to seven control and pheromone-treated plots across our study site and deployed one plant with sentinel eggs in each plot for three days (no repeated deployments in the same locations over the season)....Observations with clutches (either control or predator-exposed) that were missing or hatched were excluded, giving a final sample size of 50 (30 in the control plots vs. 20 in the pheromone-treated plots) in the analysis.

#### Lines 189–191

This adjustment for background mortality is appropriate, but it would be useful to report average non-predation egg mortality, to achieve a more realistic number attributed to predation.

**Response:** Thanks for the suggestion. The average background non-predation egg mortality was low and similar in the control and pheromone-treated plots (mean  $\pm$  SE; control plot:  $0.77 \pm 0.23$  eggs per three days; pheromone-treated plot:  $0.75 \pm 0.27$  eggs per three days). We have added this information to the methods.

**Revision:** The average non-predation mortality was low and similar in the control and pheromone-treated plots (mean  $\pm$  SE; control plot:  $0.77 \pm 0.23$  eggs per three days; pheromone-treated plot:  $0.75 \pm 0.27$  eggs per three days).

### Greenhouse Assays

#### Lines 213–214

Relative humidity (21.4%) appears unusually low for greenhouse bioassays involving predators and eggs. Please verify this value, how did the authors achieve such low humidity in the greenhouse.

**Response:** We thank the reviewer for identifying this error. Upon reviewing our records, we determined that the reported relative humidity (21.4%) resulted from a transcription error. Because we could not confidently verify the original value, we have removed the erroneous humidity measurement from the manuscript.

#### Lines 221–223

Include trial identity as a random effect if experiments were conducted sequentially.

**Response:** We thank the reviewer for this suggestion. Trial identity was included as a random effect in all greenhouse assay analyses to account for potential variation among sequential experimental trials. We apologize that this was not clearly described in the original manuscript. The Egg predation in the greenhouse section under Data analysis, has been revised to explicitly state that trial ID was included as a random effect in the generalized linear mixed-effects models.

**Revision:** Fixed effects included treatment, predator life stage/species, their interaction, and assay length as a covariate. Random effects included trial identity since experiments were conducted sequentially, and plant identity.

#### **Lines 225–229**

Sample sizes are highly unbalanced among predator species. Discuss whether statistical power differed among taxa. Also, provide justification as to why the sample size effect is not major issue in this study.

**Response:** We thank the reviewer for this comment. Sample sizes differed among predator taxa because the availability of individuals depended on successful laboratory rearing and field collections. However, control and pheromone treatments were balanced within each predator taxon (e.g., *Podisus maculiventris*: 31 control, 31 pheromone; *Lebia grandis*: 16 control, 16 pheromone; lacewing larvae: 23 control, 23 pheromone), allowing unbiased estimation of treatment effects within species. The generalized linear mixed-effects model accommodates unequal sample sizes among taxa through maximum likelihood estimation. We acknowledge that taxa represented by fewer individuals have lower statistical power and less precise estimates than taxa with larger sample sizes.

### **Statistical Analyses**

#### **Lines 236–238**

Clarify whether week was treated as a random or repeated factor. Temporal autocorrelation should be considered because plots were sampled repeatedly. Repeated measures are a standard way of analyzing this.

**Response:** We thank the reviewer for this helpful comment and apologize for the lack of clarity in the original manuscript. Week was treated as a random effect, and repeated sampling of the same experimental units was accounted for by including plot nested within row and block as random effects in the generalized linear mixed-effects model (1 | week) + (1 | block/row/plot\_ID). Thus, repeated measurements of the same plots were incorporated in the analysis. We have corrected the Statistical Analyses section to accurately describe the model structure.

**Revision:** We fit a generalized linear mixed-effects model with beetle larval abundance as the response variable, pheromone treatment as the fixed effect, with week and plot nested within the row and block as the random effects.

**Lines 245–247**

Community metrics were calculated using family-level taxa. Consider discussing limitations associated with this approach.

**Response:** We thank the reviewer for this thoughtful comment. We acknowledge that calculating community metrics at the family level may obscure species-level responses within some taxonomic groups. However, our objective was to evaluate broad community-level responses to the predator pheromone in a potato agroecosystem, where the arthropod community is relatively species-poor and dominated by a limited number of well-characterized herbivore and predator taxa. We further identified species of particular ecological importance, including *Podisus maculiventris*, *Lebia grandis*, *Coleomegilla maculata*, and the principal herbivore pests, to provide greater taxonomic resolution where biologically relevant. We have added a statement acknowledging this limitation in the statistical section.

**Revision:** Although community metrics were calculated at the family level, which may mask species-specific responses within some groups, potato arthropod communities are typically dominated by a relatively small number of herbivore and predator taxa. Consequently, we expect family-level metrics to capture the dominant community responses to pheromone deployment, while recognizing that species-level responses within families may have gone undetected.

**Lines 277–280**

The egg predation model appears to treat observations as independent. Because plots were sampled repeatedly through time, plot should likely be included as a random effect, as they are not true independent observations.

**Response:** Thanks for bringing this up. As mentioned in our previous response, we randomly selected control and pheromone-treated plots each week to deploy sentinel eggs across the study site, so there were no repeated deployments in the same locations over the season. Therefore, our observations should be largely independent and we feel that the original model specification should be appropriate.

**Results****Lines 299–302**

Please report effect size estimates (e.g., percentage reduction) in addition to statistical significance.

**Response:** We thank the reviewer for this suggestion. We have added effect-size estimates to accompany the statistical results. Estimated CPB abundance was 8.08 individuals in control plots and 5.53 individuals in pheromone-treated plots, corresponding to a 31.6% reduction in CPB abundance with pheromone exposure ( $z = 2.51$ ,  $P = 0.012$ ).

**Revised:** While the abundance of the target herbivore, the Colorado potato beetle, was reduced by the predator pheromone, none of the other herbivores were affected. The estimated mean abundance of Colorado potato beetle larvae was 32% lower in pheromone-treated plots than in control plots (5.53 vs. 8.08 larvae per plot;  $z = -2.51$ ,  $P = 0.01$ ; Fig. 1).

#### Lines 322–323

The claim of increased predation is compelling. Consider reporting estimated marginal means and confidence intervals.

**Response:** Thanks for the suggestion. We have now added the means and 95 confidence intervals to the results. We decided to report the observed means and confidence intervals instead of the estimated marginal means and the associated confidence intervals for consistency with the figure.

**Revised:** Overall egg predation rates over a two-month period were higher in the pheromone-treated plots (mean [95% CI]: 9.8 [2.7–15.9] eggs per three days) compared to the control plots (mean [95% CI]: 1.4 [0.4–2.4] eggs per three days) ( $\chi^2 = 11.7$ ;  $df = 1$ ;  $p < 0.001$ ; Fig. 4).

#### Lines 325–330

Report actual egg consumption percentages along with your statistical test results.

**Response:** We thank the reviewer for this suggestion. We have revised the Results to report actual egg consumption percentages.

**Revised:** There is no significant effect of pheromone treatment on the proportion of eggs consumed by adult *C. maculata* (81.9% vs. 82.9%;  $Z = 0.09$ ;  $p = 0.92$ ; Fig. 5), *C. maculata* larvae (85.5% vs. 68.3%;  $Z = 0.83$ ;  $p = 0.41$ ); *C. rufilabris* (20.7% vs. 13.9%;  $Z = 0.52$ ;  $p = 0.60$ ; Fig. 5), or *L. grandis* (31.1% vs. 66.1%;  $Z = -1.30$ ;  $p = 0.19$ ; Fig. 5).

There is a significant effect of pheromone treatment on the proportion of eggs consumed by *P. maculiventris* ( $Z = -3.69$ ;  $p < 0.001$ ; Fig. 5) with estimated egg consumption increasing from 1.3% in the control treatment to 29.2% in the pheromone treatment.

## Discussion

#### Lines 332–343

The discussion attributes increased field predation primarily to altered predator behavior.

Since predator activity was not directly measured, this conclusion should be presented more cautiously.

**Response:** We thank the reviewer for this comment. We agree that predator behavior was not directly measured in this study. We have revised the Discussion to present altered predator

behavior as a plausible mechanism, rather than a demonstrated explanation, for the increased egg predation observed in pheromone-treated plots.

**Revision:** Despite these neutral community responses, predation on beetle eggs was significantly higher in pheromone-treated plots, suggesting that the pheromone may have influenced predator foraging behavior rather than predator abundance. However, predator behavior was not directly measured in this study.

#### **Lines 362–372**

An important alternative explanation is that D-vac and visual sampling may not adequately detect localized changes in predator activity. So, it is always wise to discuss the study limitations.

**Response:** We thank the reviewer for this comment. We agree that D-vac and visual surveys may not detect localized or short-term changes in predator activity within plots. We have revised the Discussion to acknowledge this limitation and clarify that while we did not observe differences in predator abundance, we cannot exclude the possibility that the pheromone altered predator behavior at a finer spatial or temporal scale than our sampling methods could detect.

**Revision:** Additionally, our D-vac and visual sampling methods quantified predator abundance rather than predator activity and may not have detected localized changes in predator behavior within plots.

#### **Lines 375–380**

The proposed explanation involving increased feeding under perceived competition is interesting but speculative. Consider emphasizing that this mechanism remains untested or present it as a future direction.

**Response:** We thank the reviewer for this suggestion. We agree that the proposed mechanism is speculative. We have revised the Discussion to clarify that increased feeding under perceived competition represents a plausible hypothesis that was not directly tested in this study and should be considered a direction for future research.

**Revision:** Alternatively, increased egg consumption may reflect a behavioral response to aggregation cues that elevate local predator density, as higher conspecific densities commonly alter per-capita feeding rates through interference or competitive interactions in biological control systems (Delong & Vasseur, 2011). However, this mechanism was not directly tested in our study and warrants future investigation.

#### **Lines 397–410**

The applied implications are well presented, but the discussion would benefit from acknowledging limitations, including only one year of data, one location, one cropping system

and one pheromone applied rate (which, I had an earlier comment about how the authors selected this rate).

**Response:** We thank the reviewer for this suggestion. We agree that the applied implications should be interpreted within the scope of the study. We have revised the Discussion to acknowledge that our findings are based on a single growing season, one location, one cropping system, and a single pheromone release rate. We also note that additional studies across multiple environments and pheromone release rates are needed to evaluate the broader applicability of this approach.

**Revision:** Although these findings demonstrate the potential of predator pheromones for enhancing biological control, our conclusions are based on a single growing season at one location in a potato production system using a single pheromone release rate. Future studies evaluating multiple environments, cropping systems, years, and pheromone release rates will be important to determine the generality and optimize the application of this approach.

## Figures and Tables

### Figure 1

Include sample size and actual mean  $\pm$  SE values in either figure or caption.

**Response:** We thank the reviewer for this suggestion. We have revised the Figure 1 caption to include the raw treatment means  $\pm$  SE, along with the sample size for each treatment.

**Revised:** Figure 1: Mean ( $\pm$  SE) number of Colorado potato beetle larvae per plot in control and pheromone-treated plots averaged across the sampling period. Raw observed means were  $8.51 \pm 0.89$  larvae per plot in the control treatment ( $n = 177$  observations) and  $6.73 \pm 0.82$  larvae per plot in the pheromone treatment ( $n = 159$  observations). Error bars represent  $\pm 1$  SE. Asterisk indicates a significant treatment effect ( $p < 0.05$ ).

### Figure 3

Correct the discrepancy between the caption and text regarding predator groups shown.

**Response:** Thank you for pointing out this mistake. Caption is now correctly fixed to match predator groups shown.

**Revision:** Figure 3. Average predator number per plot across two months in control and pheromone treated plots. Predators are shown by family and are the most abundant in the field, Anthocoridae, Araneidae, Neuroptera (Chrysopidae/Hemerobiidae), Coccinellidae, Nabidae, and Pentatomidae. Error bars represent standard errors.

### Figure 4

Report exact means and confidence intervals.

**Response:** We have added the means and 95% confidence intervals to the caption.

**Revision:** Overall beetle egg predation rates in the control (mean [95% CI]: 1.4 [0.4–2.4]) and the pheromone-treated plots (mean [95% CI]: 9.8 [2.7–15.9]) over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ( $p < 0.05$ ).

### Figure 5

Consider displaying raw data points or boxplots rather than means  $\pm$  SE to illustrate variation.

**Response:** Instead of artwork, raw datapoints are now displayed on the figure.

### Minor Editorial Corrections

#### Line 75

Add a space before citation:

"(Leptinotarsa decemlineata) (Say)"

#### Line 340

Remove extra space before comma: "adults , lacewing larvae"

**Line 367** Replace: "HIPV's" with "HIPVs"

**Line 505:** Verify citation year for *emmeans* package (2025).

**Response:** Thank you for the minor editorial corrections, these have all been appropriately fixed in the main document.

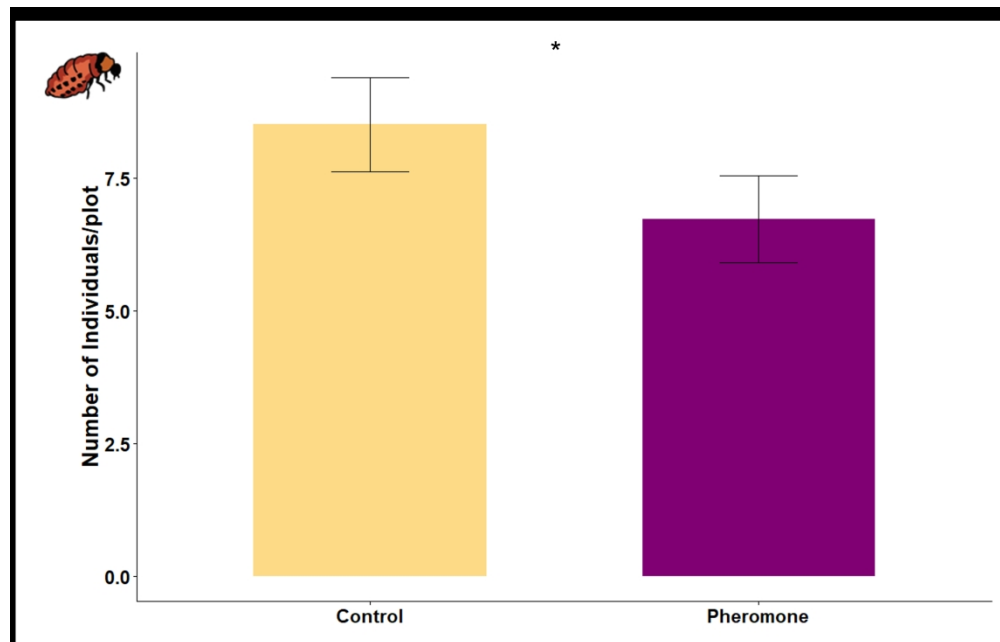


Figure 1: Number of the Colorado potato beetle larvae in the control and the pheromone-treated plots averaged over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ( $p < 0.05$ ).

1009x645mm (130 x 130 DPI)

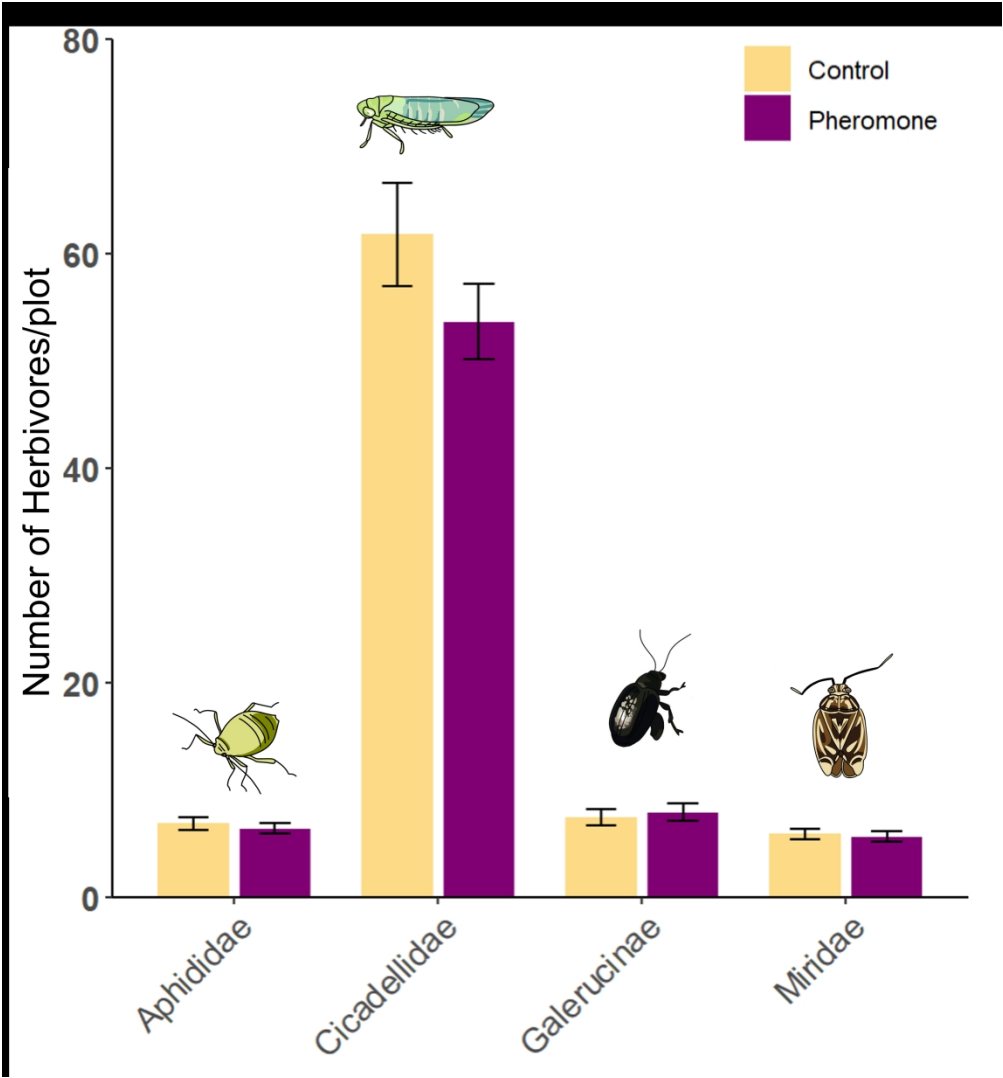


Figure 2. Average herbivore number per plot across two months in control and pheromone-treated plots. Herbivores are shown by family and were the major herbivore families found in the field. Aphididae (aphids), Cicadellidae (potato leafhopper), Galerucinae (flea beetles), and Miridae (tarnished plant bugs). Error bars represent standard errors.

682x733mm (130 x 130 DPI)

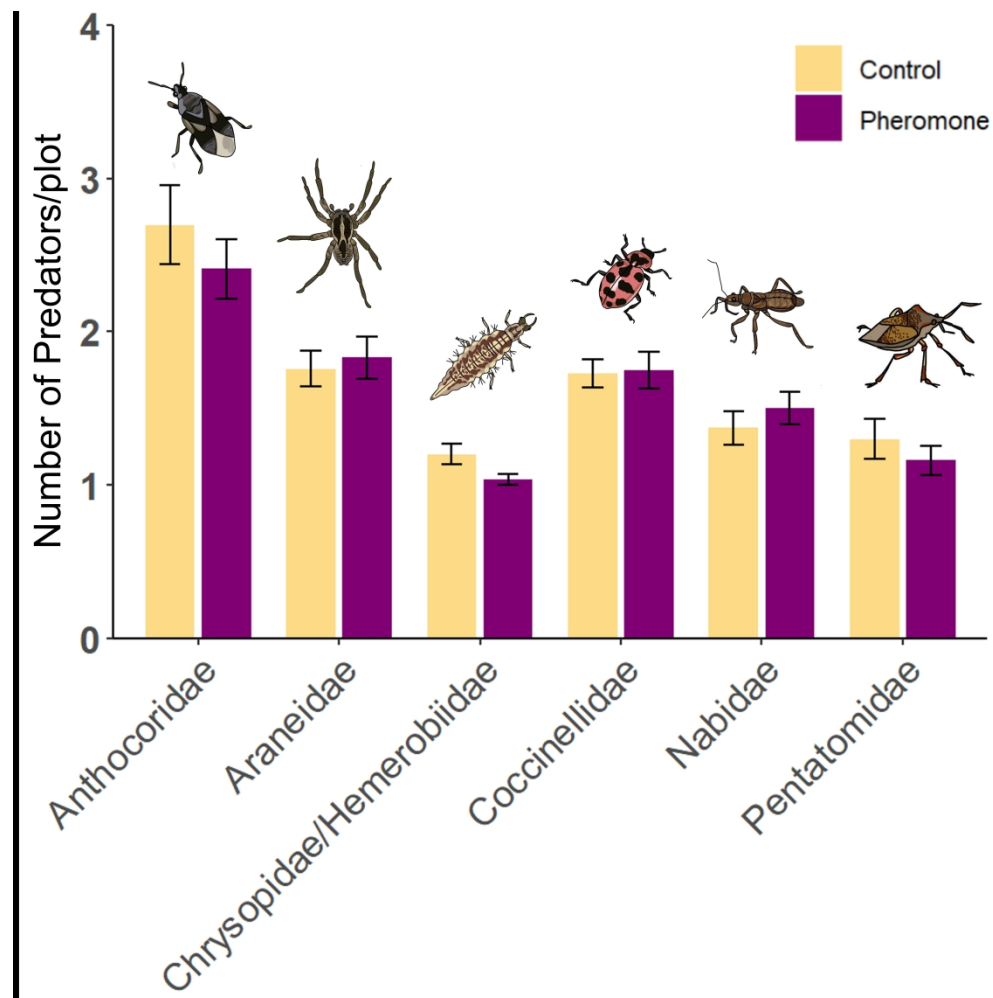


Figure 3. Average predator number per plot across two months in control and pheromone treated plots. Predators are shown by family and are the most abundant in the field, Anthocoridae, Araneidae, Coccinellidae, Nabidae, and Neuroptera. Error bars represent standard errors.

676x670mm (130 x 130 DPI)

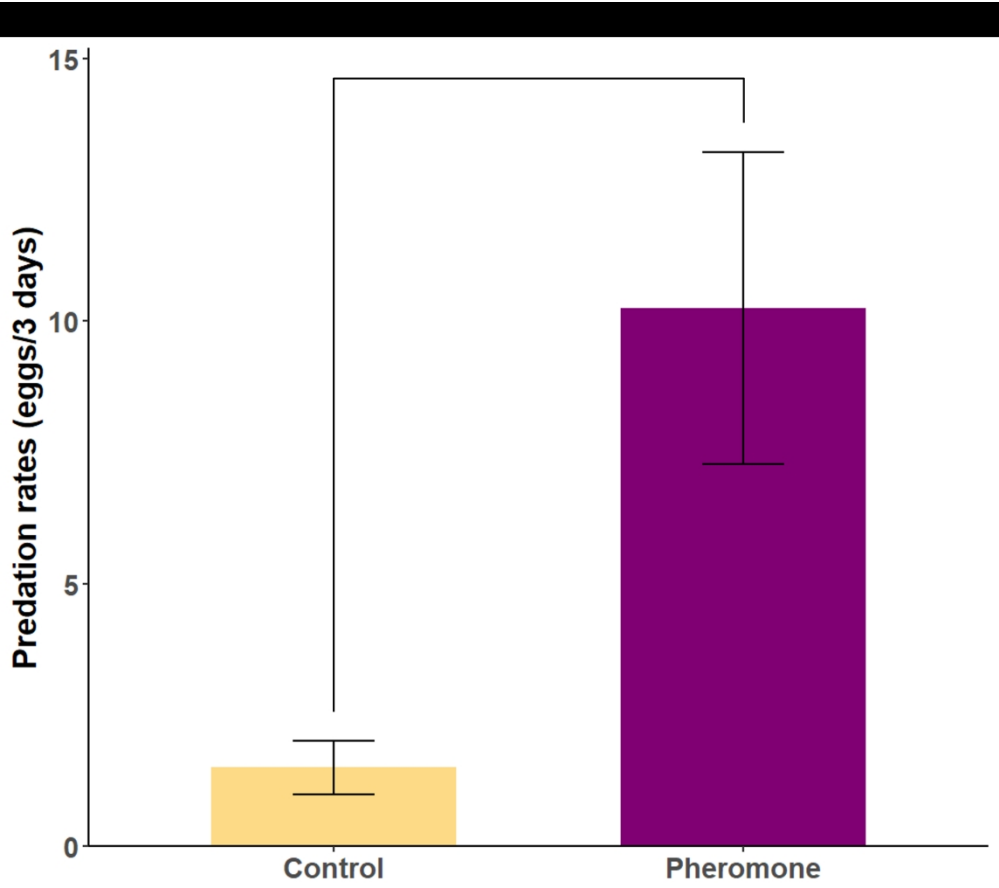


Figure 4. Overall beetle egg predation rates in the control and the pheromone-treated plots over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ( $p < 0.05$ ).

741x681mm (130 x 130 DPI)

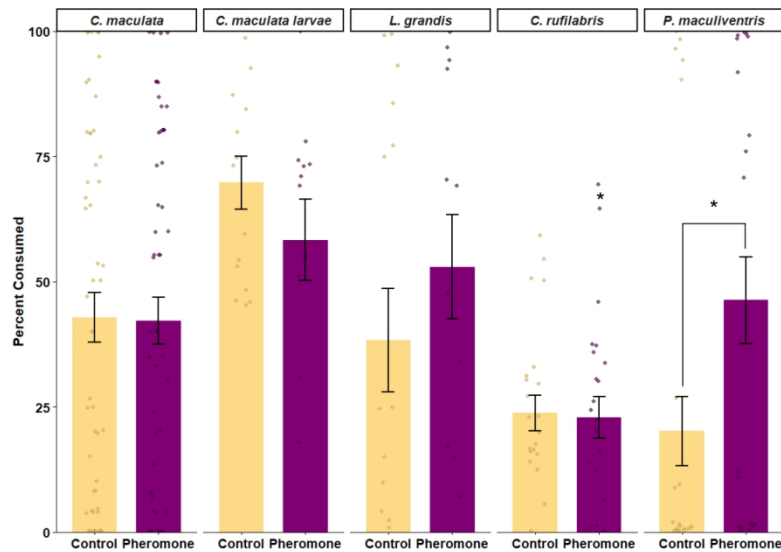
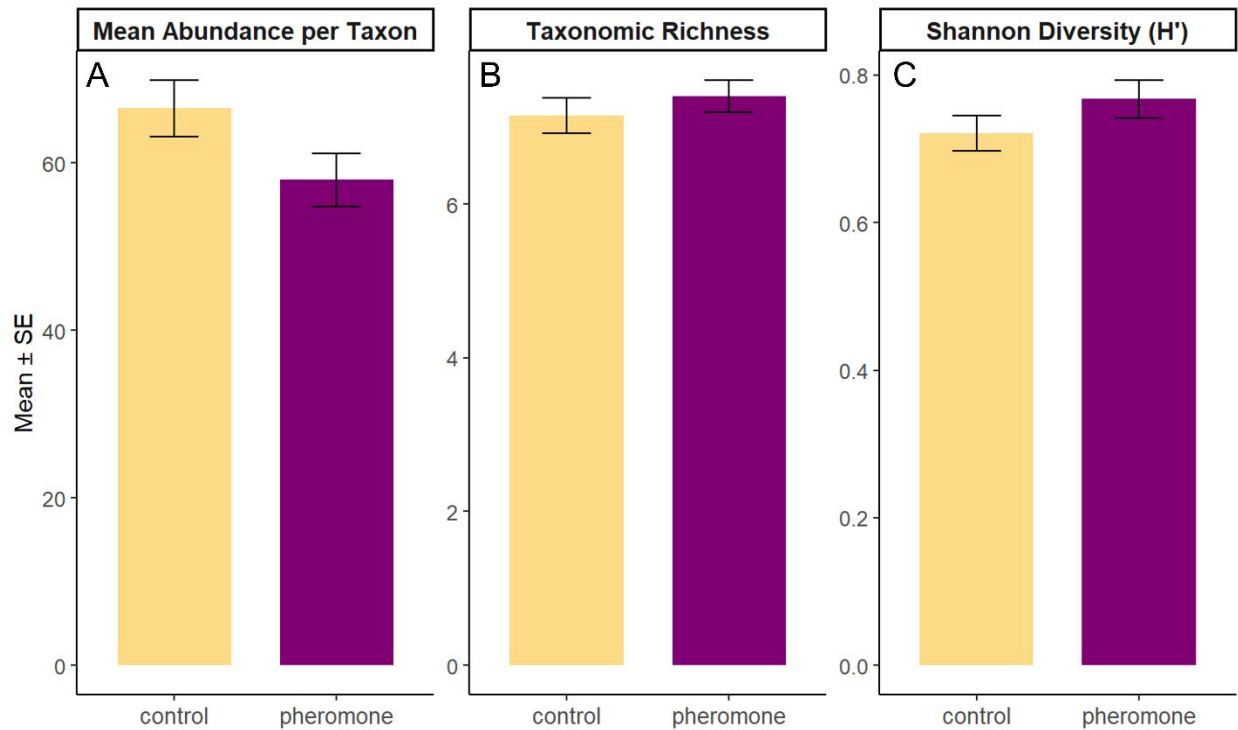


Figure 5. Percent of the Colorado potato beetle eggs consumed by five predator species/life stages when exposed to predator pheromone or control volatiles in greenhouse assays. Panels from left to right show *Coleomegilla maculata* adults, *Chrysoperla rufilabris*, *Lebia grandis*, *Podisus maculiventris*, and *Coleomegilla maculata* larvae. Error bars represent standard errors. Asterisks denote significant differences between treatment levels ( $p < 0.05$ ).

677x381mm (72 x 72 DPI)

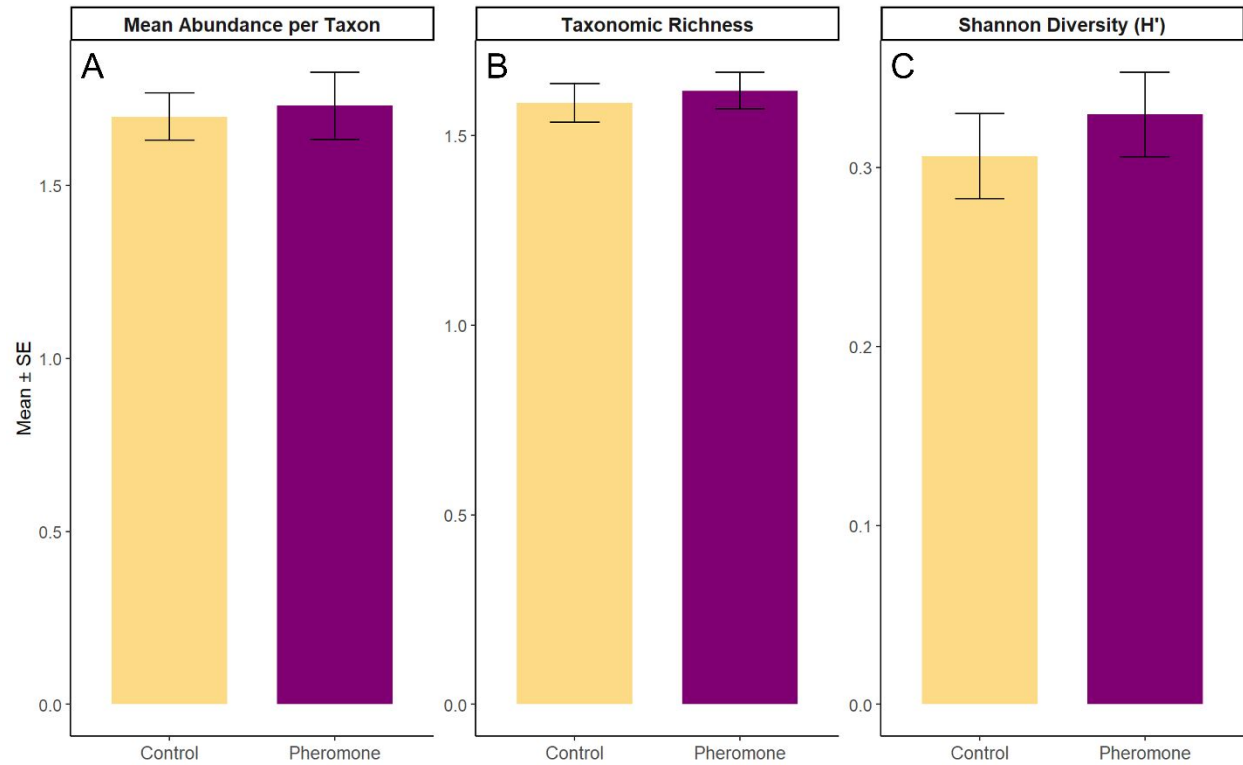
Supplemental/extra figures:



**Supplemental Figure S1.** A) Mean abundance, B) Richness, C) Diversity of herbivore taxa in the control and the pheromone-treated plots over a two-month period. Error bars represent standard errors.

**Table S1.** Statistical results examining the effects of pheromone treatment on herbivore abundance, richness, and diversity.

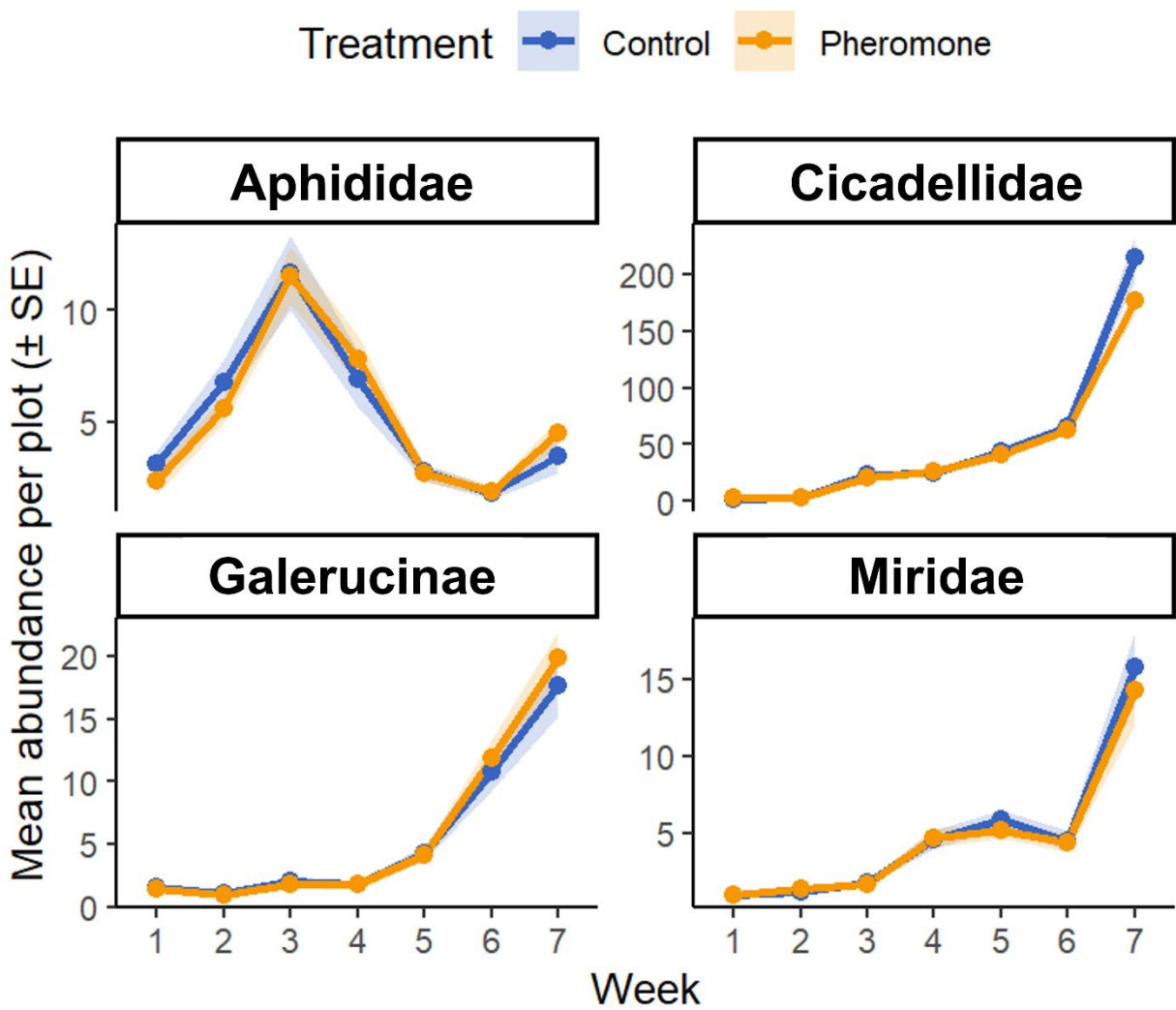
| Response variable | Test statistic   | df    | <i>p</i> -value |
|-------------------|------------------|-------|-----------------|
| <i>Abundance</i>  | <i>t</i> = −0.91 | 139.5 | 0.36            |
| <i>Richness</i>   | <i>Z</i> = −0.22 | ---   | 0.83            |
| <i>Diversity</i>  | <i>t</i> = 0.36  | 139.9 | 0.72            |



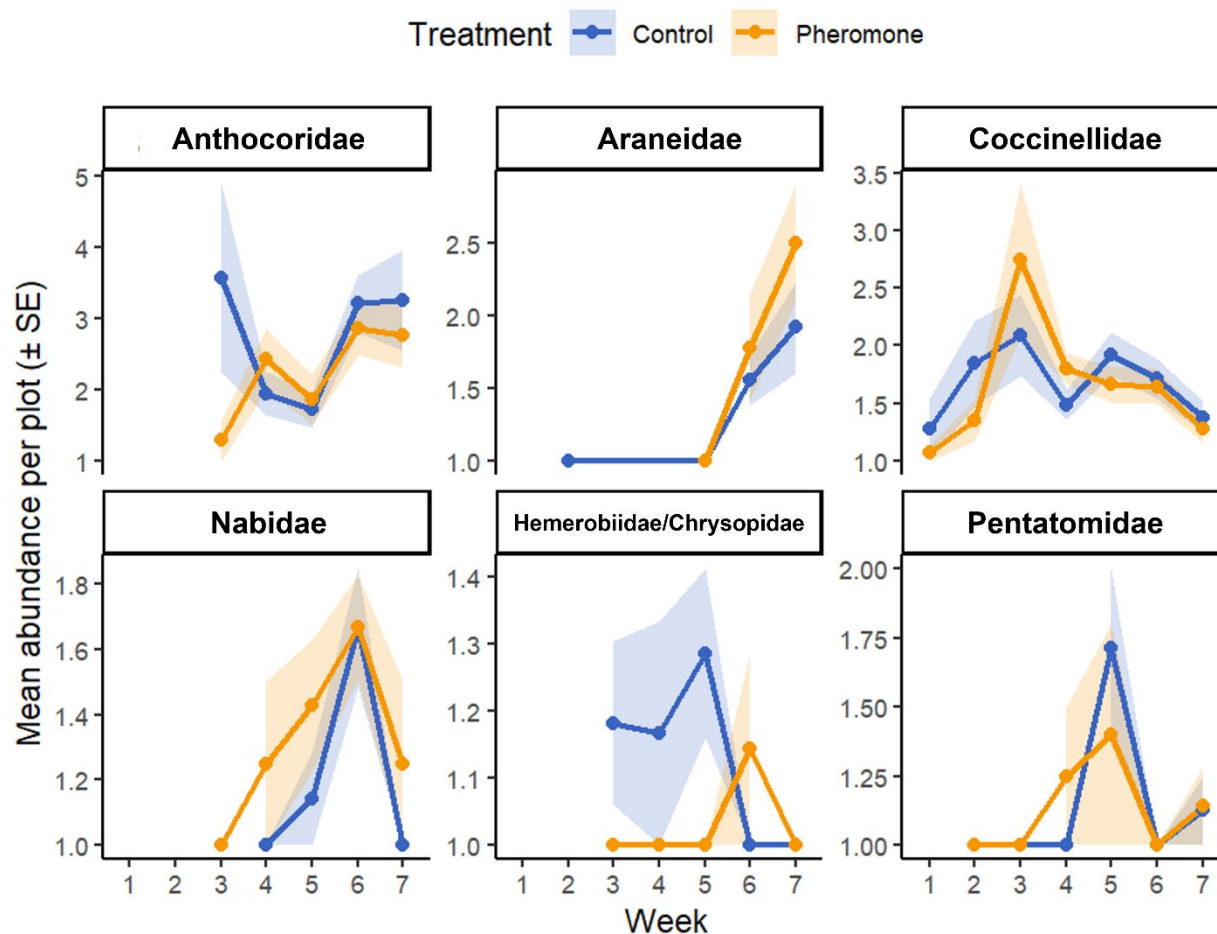
**Supplemental Figure S2.** A) Mean abundance, B) Richness, C) Diversity of predator taxa in the control and the pheromone-treated plots over a two-month period. Error bars represent standard errors.

**Table S2.** Statistical results examining the effects of pheromone treatment on predator abundance, richness, and diversity.

| Response variable | Test statistic | df  | <i>p</i> -value |
|-------------------|----------------|-----|-----------------|
| <i>Abundance</i>  | $t = -0.12$    | 124 | 0.90            |
| <i>Richness</i>   | $Z = 0.51$     | --- | 0.61            |
| <i>Diversity</i>  | $t = 0.84$     | 134 | 0.40            |



**Supplemental Figure S3.** Pest abundance shown over the seven-week period. Each line represents the average abundance of herbivores per plot over time. The blue line represents control treatment, and the orange line represents the pheromone treatment.



**Supplemental Figure S4.** Predator abundance shown over the seven-week period. Each line represents the average abundance of predators per plot over time. The blue line represents control treatment, and the orange line represents the pheromone treatment.